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Galactic cirrus in the Perseus Cluster field

A panchromatic view with *Euclid*, CFHT, GALEX, IRIS, and *Planck*

Qing Liu[1⋆], Henk Hoekstra[1], Jean-Charles Cuillandre[2], Marc-Antoine Miville-Deschênes[3], Matthias Kluge[4], Nina A. Hatch[5], René Laureijs[6], Amaël Ellien[7], Petra Awad[1], Juan Miró-Carretero[8,1] and David Valls-Gabaud[9]

[1] Leiden Observatory, Leiden University, P.O. Box 9513, 2300 RA Leiden, The Netherlands
[2] Université Paris-Saclay, Université Paris Cité, CEA, CNRS, AIM, 91191, Gif-sur-Yvette, France
[3] Laboratoire de Physique de l'Ecole normale supérieure, ENS, Université PSL, CNRS, Sorbonne Université, Université Paris-Diderot, Sorbonne Paris Cité, Observatoire de Paris, Paris, France
[4] Max Planck Institute for Extraterrestrial Physics, Giessenbachstrasse, D-85748 Garching, Germany
[5] School of Physics and Astronomy, University of Nottingham, University Park, Nottingham NG7 2RD, UK
[6] Kapteyn Astronomical Institute, University of Groningen, PO Box 800, 9700 AV Groningen, The Netherlands
[7] OCA, P.H.C Boulevard de l'Observatoire CS 34229, 06304 Nice Cedex 4, France
[8] Departamento de Física de la Tierra y Astrofísica, Universidad Complutense de Madrid, E-28040 Madrid, Spain
[9] LUX, Observatoire de Paris, CNRS, PSL, 61 Avenue de l'Observatoire, 75014 Paris, France

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ABSTRACT

Galactic cirrus is both a foreground for low surface brightness extragalactic science and a resolved tracer of dust scattering in the diffuse interstellar medium. We study Galactic cirrus using the *Euclid* Early Release Observation of the Perseus cluster, where prominent cirrus is visible across the field. We combine *Euclid* VIS and NISP imaging with CFHT *ugriz*, GALEX FUV and NUV, IRIS 100 μm data, and *Planck* thermal dust products to measure the diffuse Galactic light (DGL) from UV to NIR. The DGL is measured from its spatial correlation with far-infrared dust emission and explained with a radiative transfer model for a plane-parallel Galactic disk. The Perseus DGL SED rises from the UV to a broad peak in the optical and declines at NIR wavelengths, consistent with dust scattered Galactic starlight. The $H_{\rm E}$ intensity has an excess relative to the scattered light model, suggesting a contribution from small grain emission. We use Bayesian inference to constrain the scattering parameters. We find the albedo is well constrained, with $\omega \simeq 0.42$–0.47 in the FUV, $\omega \simeq 0.56$–0.65 across the optical bands, and $\omega \simeq 0.5$–0.6 in the NIR. The inferred scattering asymmetry parameter g remains prior-driven, suggesting that broadband photometry alone from a single sightline is not sufficient to determine the scattering phase function. The optical-to-NIR colors provide a useful diagnostic for distinguishing diffuse extragalactic stellar emission from cirrus. The clearest separation based on the Perseus data is found using $I_{\rm E} - J_{\rm E}$ versus $g - r$. These results show that cirrus in deep imaging is not only a foreground systematic, but also can be a valuable source of information about the diffuse ISM.

Key words. ISM: dust, extinction – radiative transfer – galaxies: clusters: individual: Perseus – galaxies: photometry – techniques: image processing

1. Introduction

Galactic cirrus, **first identified as filamentary emission in far-infrared (FIR) observations** (Low et al. 1984), traces diffuse dust heated and illuminated by the Galactic starlight. At ultraviolet (UV), optical, and near-infrared (NIR) wavelengths, cirrus is observed primarily as diffuse Galactic light (DGL), produced when starlight is scattered into the line of sight by dust grains in the diffuse interstellar medium (ISM; e.g., Elvey & Roach 1937; Henyey & Greenstein 1941; Mattila 1979; Draine 2003b). The brightness and spectral shape of the DGL depend jointly on the illuminating interstellar radiation field (ISRF), the dust column density, and the scattering properties of the grains (Gordon et al. 1994; Draine 2003b). Measurements from UV through NIR have therefore been used to constrain dust albedo, scattering asymmetry, grain size distributions, and **to identify emission other than scattered light.** (e.g., Witt et al. 1997; Brandt & Draine 2012; Ienaka et al. 2013; Arai et al. 2015; Sano et al. 2015; Mattila et al. 2018, 2023). DGL is also strongly correlated with FIR thermal emission of the same dust, providing an independent tracer of dust column density and heating (Laureijs et al. 1987; Guhathakurta & Tyson 1989; Brandt & Draine 2012; Ienaka et al. 2013; Onishi et al. 2018; Zhang et al. 2023). DGL therefore provides a complementary probe to constrain dust properties in addition to the information obtained from extinction, polarization, and thermal infrared emission (Draine 2003a), while retaining information about the geometry and the illuminating ISRF.

The major difficulty in studying Galactic cirrus is its extremely low surface brightness. At high Galactic latitude, optical cirrus is typically orders of magnitude fainter than the night sky, placing it in the same observational regime as diffuse stellar halos, tidal debris, intracluster light (ICL), and other low surface brightness structures (Duc et al. 2015; Mihos et al. 2017; Román et al. 2020). Preserving these faint signals requires careful control of flat-fielding, sky subtraction, instrumental scattered light, and the extended wings of the point-spread function (PSF; Slater et al. 2009; Sandin 2014; Fliri & Trujillo 2016; Liu et al. 2022, 2023; Watkins et al. 2024; Cuillandre et al. 2025b). Conventional

⋆ e-mail: qliu@strw.leidenuniv.nl

reductions optimized for compact-source science can suppress or distort the diffuse emission of interest. Recent advances in low surface brightness instrumentation and data processing have greatly improved our ability to preserve such diffuse emission. Deep wide-field imaging provides **multiband photometry** of cirrus and its relation to FIR dust emission (Román et al. 2020; Smirnov et al. 2023; Zhao et al. 2024; Liu et al. 2025a,b). Optical imaging also reveals structures on substantially finer angular scales than conventional dust probes (Miville-Deschênes et al. 2016; Marchuk et al. 2021; Liu et al. 2025b). Optical Galactic cirrus has therefore become accessible not only as a diffuse foreground, but also as a spatially resolved tracer of the multi-scale ISM.

The same complex morphology that make cirrus scientifically interesting also pose a challenge for low surface brightness extragalactic science. At the surface brightness levels where stellar halos, tidal streams, and ICL become visible, Galactic cirrus can appear as faint blobs, broad sheets, narrow filaments, and other coherent structures that closely resemble features produced by genuine extragalactic emission (Smirnov et al. 2023; Liu et al. 2025a,b; Poliakov et al. 2026). Cirrus can also have optical colors comparable to those of diffuse stellar populations (Román et al. 2020; Ragusa et al. 2021; Smirnov et al. 2023; Zhao et al. 2024; Liu et al. 2025a). **Observations of the Perseus cluster offer a classic example** of this problem. The *Euclid* Early Release Observation (ERO) reveals an extended halo of the brightest cluster galaxy (BCG) NGC 1275 and a surrounding ICL component in the cluster. However, prominent Galactic cirrus is also detected across the field at comparable surface brightness levels (Cuillandre et al. 2025a; Kluge et al. 2025). As a result, cirrus may affect measurements of the morphology, color, and radial extent of the ICL (Ellien et al. 2025). The BCG, ICL, or extended galaxy halos can in turn contaminate measurements of the cirrus itself. Reliable low surface brightness science in such fields therefore requires the Galactic and extragalactic diffuse components to be characterized simultaneously.

The *Euclid* space telescope offers an unprecedentedly powerful window for studying Galactic cirrus. With its excellent control in both compact and diffuse emission observations, the VIS imaging provides a deep, highly resolved view of cirrus in the optical, while the NISP bands extend this view into the underexplored NIR regime (Euclid Collaboration: Scaramella et al. 2022; Euclid Collaboration: Mellier et al. 2025). Most spatially resolved DGL studies have focused on the optical. NIR DGL has been measured by satellites and rocket experiments, including COBE/DIRBE, CIBER, and MIRIS, but generally at much coarser angular resolution (Arai et al. 2015; Sano et al. 2015; Onishi et al. 2018). *Euclid* can instead probe the same diffuse structures from the optical into the NIR at high angular resolution, providing spectral energy distribution (SED) measurements of cirrus over a broad wavelength coverage. The Perseus ERO field is particularly useful in this context because it combines prominent Galactic cirrus with a rich cluster environment containing the extended halo of NGC 1275, ICL, and other diffuse galaxies (Cuillandre et al. 2025b). It therefore provides an opportunity both to characterize DGL and to investigate how Galactic cirrus can be distinguished from faint extragalactic structures in *Euclid* imaging.

A better characterization of DGL is also important for measurements of the diffuse extragalactic background. In the infrared, Galactic dust emission is one of the major foreground components that must be separated from the cosmic infrared background (CIB; Hauser et al. 1998; Arendt et al. 1998; Auclair et al. 2024; Sinha et al. 2026). In the optical, DGL similarly contributes to measurements of the cosmic optical background (COB; Matsuoka et al. 2011; Zemcov et al. 2017; Lauer et al. 2021; Postman et al. 2024). The need to characterize cirrus/DGL will extend to other wide-field facilities. The Nancy Grace Roman Space Telescope will provide deep, high-resolution NIR imaging over wide areas of the sky (Akeson et al. 2019), while SPHEREx provides complementary all-sky spectrophotometric coverage in the NIR (Doré et al. 2014; Crill et al. 2020). A physically motivated characterization of DGL across these wavelengths is therefore relevant both to studies of the diffuse ISM and to measurements of faint extragalactic emission.

In this work, we characterize the DGL in the Perseus field from the UV to the NIR using complementary datasets and interpret its SED in terms of dust scattering. We ask three connected questions: how does the cirrus emission change from the UV to the NIR, whether its measured SED can be described by a physical dust scattering model, and what constraints can such measurements place on the scattering properties of interstellar dust. We give extra attention to separating the Galactic foreground light from extragalactic emission, including the BCG, ICL, and galaxy halos. We then compare the measurements with a radiative transfer model and use the optical–NIR colors to assess how effectively cirrus can be distinguished from extragalactic low surface brightness features in the same imaging data. We use a Bayesian framework to infer the dust scattering parameters. The paper is organized as follows. Sec. 2 describes the datasets and post-processing procedures. Sec. 3 presents the DGL measurement, and Sec. 4 introduces the radiative transfer model. Sec. 5 describes the inference framework. Sec. 6 presents the inference results, and Sec. 7 discusses their implications and systematics in our measurement and modeling.

2. Observations and data processing

This section describes the multi-wavelength data set used to measure the DGL in the Perseus field. The data set includes *Euclid* VIS and NISP imaging, CFHT/MegaCam (*ugriz*) imaging, GALEX FUV and NUV observations, and thermal dust emission maps from IRAS and Planck. Figure 1 displays the RGB composite of the Perseus field using CFHT and *Euclid* data.

2.1. Euclid ERO VIS and NISP imaging

We use the *Euclid* ERO imaging (Euclid Early Release Observations 2024) of the Perseus cluster in the broad VIS (I_E) band and the NISP (Y_E, J_E, and H_E) bands. The ERO observations and reduction pipeline are described in Cuillandre et al. (2025b), with an overview of the Perseus dataset presented in Cuillandre et al. (2025a). The Perseus observations combine four reference observation sequences and are deeper than a Euclid Wide Survey (EWS) field. The ERO pipeline was developed to retain both compact sources and faint extended emission. It corrects the principal instrumental signatures in the VIS and NISP exposures, performs astrometric and photometric calibration, and combines individual exposures into science stacks. For this work, we use the stacks optimized for diffuse emission. VIS has a native pixel scale of 0.″1, while the NISP images have a pixel scale of 0.″3. We adopt the ERO photometric calibration from Cuillandre et al. (2025b) and convert the calibrated images to specific intensity in kJy/sr. Further processing of the diffuse emission, including the removal of faint blobs, the BCG and ICL, instrumental residuals, and zodiacal light, **is described in Sec. 2.5 and Appendix A**.

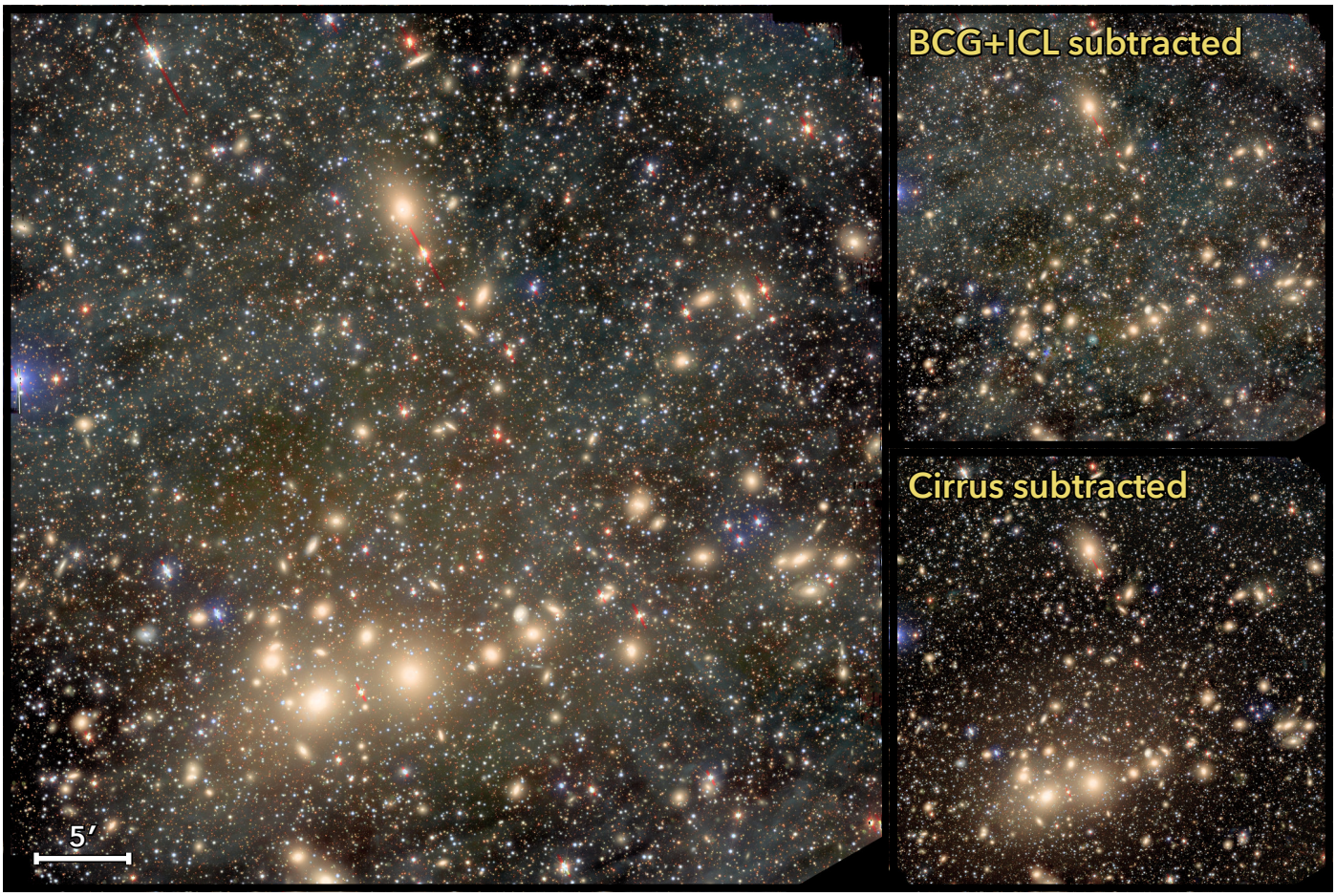


Fig. 1: **Left**: Color composite of the Perseus field. *B*: CFHT *u*, *g*; *G*: CFHT *r*, *i* + *Euclid* I_E; *R*: *Euclid* Y_E, J_E, and H_E. The faint ICL and foreground Galactic cirrus is prominent as diffuse light in the sky background. **Upper right**: Color composite of BCG+ICL-subtracted Perseus field. The second brightest galaxy NGC1272 is also subtracted. **Lower right**: Color composite of cirrus-subtracted Perseus field. The cirrus-subtracted images are generated using the ICL models and cirrus maps.

2.2. CFHT *ugriz* imaging

We complement the *Euclid* data with deep *ugriz* imaging obtained with MegaCam on the Canada-France-Hawaii Telescope (CFHT). The CFHT observations of the Perseus field are described in Cuillandre et al. (2025a) and Mondelin et al. (2026). The images were processed with the Elixir-LSB pipeline (Ferrarese et al. 2012), which was designed to control large-scale instrumental backgrounds while retaining faint extended emission. MegaCam has a native pixel scale of 0.″187. The image quality varies between bands, with seeing ranging from 0.″7 to 1.″4. The final stacks use a common AB magnitude zero point of 30. We converted the calibrated images to kJy/sr before applying the subsequent source removal, diffuse light separation, and background corrections. The CFHT data are **post-processed and measured with the same procedures as the *Euclid* data so that the optical and NIR measurements are directly comparable**.

2.3. IRAS and Planck dust tracers

We use two complementary tracers of thermal dust emission: the 100 μm intensity from IRAS and the dust radiance from the *Planck* all-sky thermal dust model (Planck Collaboration 2014b). The 100 μm map is taken from the Improved Reprocessing of the IRAS Survey (IRIS; Miville-Deschênes & Lagache 2005), which improves the calibration, destriping, and zodiacal-light subtraction of the original IRAS products. We denote the 100 μm intensity by I_{100}. Although the angular resolution (4.3′) is much coarser than that of the optical and NIR imaging, the 100 μm intensity is the conventional FIR normalization for DGL measurements.

The IRIS 100 μm intensity contains an approximately isotropic contribution from the cosmic infrared background (CIB). We estimated this contribution using the *Planck* all-sky thermal dust model, following the approach used in Liu et al. (2025a). The *Planck* model fits the thermal dust SED using the high-frequency *Planck* measurements and IRAS data, providing maps of dust temperature, optical depth, spectral index, and radiance. From the fitted thermal model, we predicted the Galactic dust emission at 100 μm and compared it with the observed IRIS map. The 2D difference between the two maps was removed from IRIS, yielding the corrected I_{100} map used in this work.

We also use the *Planck* dust radiance, $\mathcal{R}$, defined as the integrated thermal dust emission power. In contrast to a single-band FIR intensity, the radiance incorporates the dust emission over the full thermal spectrum and is less directly modulated by variations in dust temperature. We therefore use $\mathcal{R}$ as the main dust emission tracer when measuring the DGL to bridge between I_{100} and the optical-NIR intensity.

In addition, we use the *Planck* 2013 High Frequency Instrument (HFI) maps at 353, 545, and 857 GHz as **consistency** checks of the long-wavelength thermal dust SED. We adopt the HFI products with zodiacal emission and stray light corrected (Planck Collaboration 2014a,c). They trace the submillimeter emission from the large-grain dust population, sampling the long wavelength thermal tail.

2.4. GALEX UV imaging

We include GALEX FUV and NUV imaging to extend the DGL measurement into the UV, where the wavelength dependence of dust scattering and the illuminating radiation field differs strongly from that in the optical and NIR. The GALEX observations do not cover the full *Euclid* Perseus field; the usable FUV/NUV regions with high relative response extend over approximately 0.6°/0.45° **in radius** around the BCG, respectively.

We constructed the UV maps from GALEX products downloaded from the Mikulski Archive for Space Telescopes (Morrissey et al. 2007). The mosaics were processed on the supplied 0.″6 grid and converted from count-rate units to physical surface brightness units kJy/sr using the conversion factors from the GALEX calibration. The diffuse UV sky observed by GALEX contains foreground emission from airglow and zodiacal light. Their removal is described in Appendix A.3.

2.5. Processing of diffuse emission

Before measuring the DGL, we processed the data set to extract diffuse emission in the images. The procedure differs slightly between datasets. **The detailed processing to extract the diffuse emission, including compact emission removal, morphological filtering, foreground subtraction, BCG+ICL modeling, and halo removal is described in Appendix A.**

The *Euclid* and CFHT images were first processed using the `DeNeb` software (Bouy et al. 2025; Bertin et al., in prep.) to separate compact sources from diffuse emission, followed by additional cleaning of localized residuals such as bright galaxy halos, stellar ghosts, and artifacts **through blob detection and masking**. Zodiacal light was removed from the optical and NIR data using the `ZodiSURF` model (O'Brien et al. 2026). For GALEX, we instead removed airglow from both FUV and NUV and zodiacal light from NUV using the diffuse-background estimates of Murthy (2014).

In the optical and NIR, we further subtracted the BCG+ICL component and modeled or masked the extended halos of bright stars and galaxies, optical ghosts, as well as remaining detector artifacts. No ICL subtraction was applied to the GALEX images, where the diffuse stellar emission is expected to be negligible. The BCG NGC 1275 and its surroundings were conservatively masked in GALEX images. The resulting masks were filled in for continuous support required for image convolution, which is described in Appendix B.

Figure 2 shows the resultant cirrus maps of the Perseus field in the CFHT/MegaCam *ugriz*, the *Euclid* $I_E/Y_E/J_E/H_E$, and the GALEX FUV/NUV bands. The IRIS 100 μm map and the *Planck* radiance map are also shown.

3. Measurement of the DGL SED

3.1. Measurement preparation

*skew*Before measurement, all bands were converted to a common surface brightness unit of kJy/sr. We describe the DGL SED using the band-averaged intensity normalized by the 100 μm emission,

$$y_\nu = \frac{\langle \nu I_\nu \rangle}{I_{100}}, \tag{1}$$

where $\langle \nu I_\nu \rangle$ is the bandpass-averaged spectral surface brightness.

We reprojected the *Euclid*, CFHT, GALEX, *Planck* radiance, and IRIS 100 μm maps on a common grid with a pixel scale of 0.″6. The higher-resolution images were convolved to match the beam of the FIR data using wrapped boundary conditions. We restricted the analysis to pixels above the lowest 1% of the *Planck* radiance distribution. This removes the lowest-dust regions of the field, where the correlation is weak and more sensitive to residual background structure.

Before measuring the correlations, we independently subtracted a linear spatial plane from the *Planck* radiance and IRIS maps. The planes were fitted with three iterations of 3σ clipping. This removes broad map-scale gradients while retaining the smaller-scale dust structure that drives the DGL correlation.

3.2. Correlation with far-infrared dust emission

The DGL intensity was measured by correlating the surface brightness in each band with the dust thermal emission. At frequency ν, we modeled the observed intensity as

$$I_\nu(\mathbf{x}) = \alpha_\nu \, I_{\rm dust}(\mathbf{x}) + P_\nu(\mathbf{x}) + \epsilon_\nu(\mathbf{x}), \tag{2}$$

where $I_{\rm dust}(\mathbf{x})$ is the dust tracer (here the *Planck* dust radiance $\mathcal{R}$), α_ν is the band-dependent dust-correlation slope, $P_\nu(\mathbf{x})$ describes a smooth residual background, and $\epsilon_\nu(\mathbf{x})$ contains noise and remaining residuals. The zero-point offset of each image was not constrained and was absorbed by $P_\nu(\mathbf{x})$. We used a second-order polynomial for $P_\nu(\mathbf{x})$, which allows for large-scale residual backgrounds. The slopes and polynomial coefficients were estimated simultaneously for data from each facility using penalized least square minimization applied to the polynomial terms. This reduces degeneracy between the dust-correlation signal and the band-dependent residual backgrounds.

To express the measurement relative to 100 μm emission, we fit the relation

$$I_{100}(\mathbf{x}) = \alpha_{100} \, \mathcal{R}(\mathbf{x}) + \epsilon_{100}(\mathbf{x}) \tag{3}$$

between the *Planck* radiance map and the IRIS intensity map. Because large-scale planes have already been removed from both FIR maps, no additional spatial nuisance term is included in this fit. The resulting DGL SED is $y_\nu = \langle \nu I_\nu \rangle / I_{100} = \nu_{\rm eff} \, \alpha_\nu / \alpha_{100}$, where $\nu_{\rm eff}$ is the effective frequency of the corresponding band. As a result, all UV, optical, and NIR measurements were placed on the same normalization through their correlation with the dust thermal emission.

3.3. Jackknife covariance and uncertainties

We estimated the uncertainty and inter-band covariance of the Perseus DGL SED with a paired spatial jackknife. The common footprint was divided into a 5×5 grid. In each realization, one region was removed from every band and from both FIR maps, and the measurement was repeated. For each jackknife realization, we refit the planes in the *Planck* and IRIS maps and recomputed the correlations. Repeating the analysis with 4×4 and 6×6 grids gives comparable results.

The jackknife measures how strongly the inferred SED depends on the spatial structure sampled within the Perseus field.

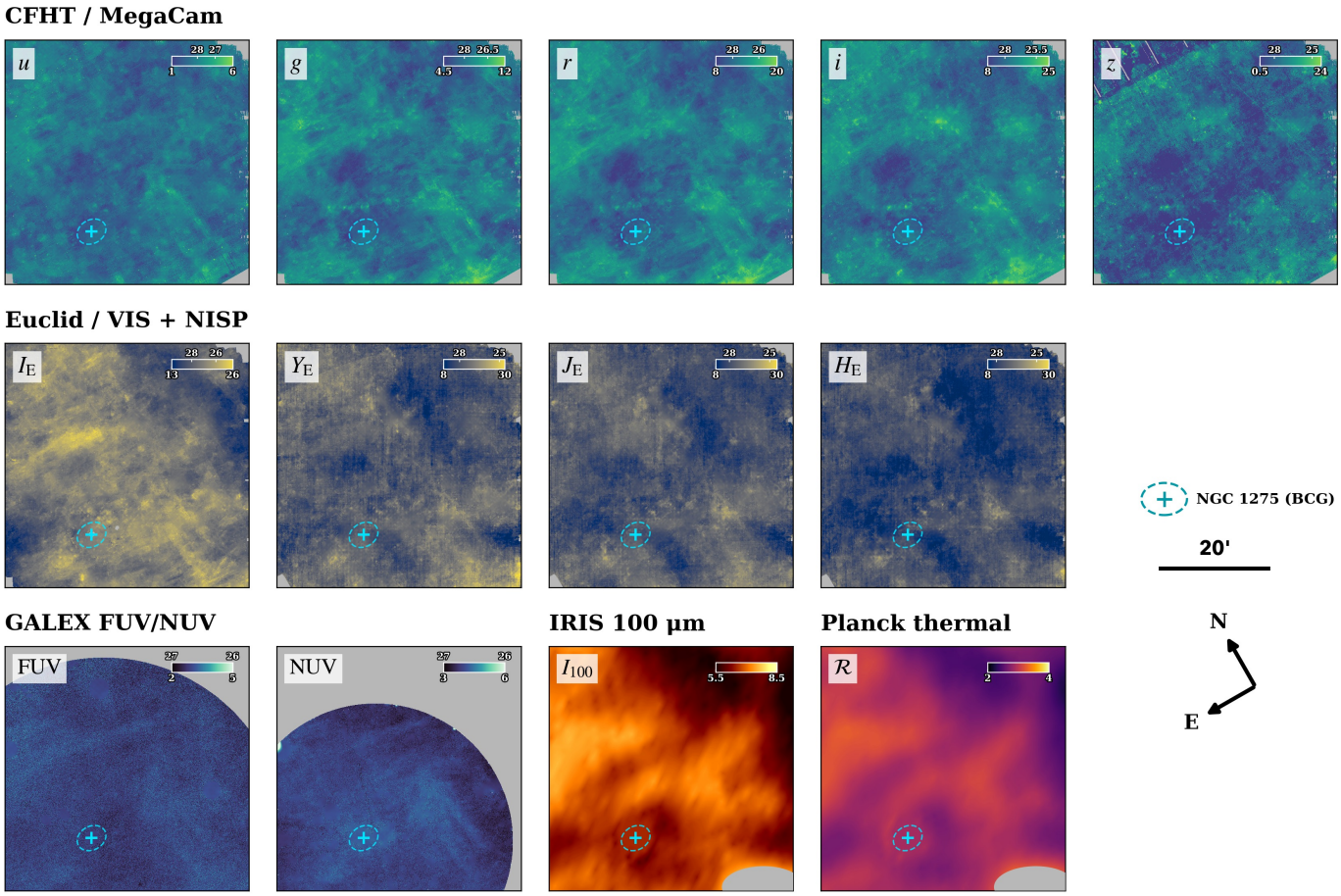


Fig. 2: Multiwavelength view of the Galactic cirrus in the Perseus field. **Upper row**: the CFHT/MegaCam *ugriz* maps. **Middle row**: the *Euclid* VIS and NISP maps. **Bottom row**: the GALEX FUV/NUV maps, the IRIS 100 μm map, and the *Planck* radiance map. For UV to NIR maps, the color bar shows the intensity in kJy/sr and the surface brightness in mag/arcsec^2 with reference to a band-dependent background sky level corresponding to the 1% quantile of *Planck* dust radiance. The unit is in MJy/sr for IRIS map and in 10^{-7}W/m^2/sr for the *Planck* map. The cyan cross marks the BCG (NGC 1275), and the dashed cyan ellipse marks a semi-major-axis radius of 60 kpc, roughly corresponding to the BCG and ICL transition. The bright source in the bottom right corner of IRIS and *Planck* maps is masked.

It captures uncertainties in the sampling of individual regions, the dynamic range of dust-emission, the fitted large-scale gradients, the FIR normalization, and the band-dependent residual background.

The full covariance was retained for the subsequent radiative transfer inference. By construction, the inter-band covariance terms between different facilities were set to zero. These uncertainties do not include the field-to-field variation of DGL across the sky, which would require multiple sightlines. [1] We report the paired jackknife as the nominal uncertainty estimate. Systematic uncertainties from the ICL and zodiacal light removal were not included in the jackknife, and we discuss them in Sec. 7.2.

4. Radiative transfer model

We interpret the measured DGL SED with a single-scattering radiative transfer model in an infinite plane-parallel Galactic disk, following Brandt & Draine (2012) and Chellew et al. (2022). The Sun is placed at the Galactic midplane. The stellar emissivity follows two exponential disks with scale heights of 300 and 1350 pc, with 90% of the luminosity in the thinner component. Dust follows a Gaussian vertical distribution with a scale width of 250 pc. Its normalization is set by a one-sided vertical optical depth of $\tau_{V,0} = 0.15$ from the midplane to $z = +\infty$, giving a line-of-sight optical depth of $\tau_{V,0} \csc|b|$ at Galactic latitude b.

We adopt the local ISRF of Mathis et al. (1983), hereafter MMP83, as the illuminating spectrum. For the adopted stellar vertical distribution, the intrinsic stellar emissivity is normalized such that the attenuated radiation field at the Galactic midplane reproduces the MMP83 ISRF.

For the nominal dust model, we use the updated optical properties tabulated by Draine (2003b), based on the Milky Way $R_V = 3.1$ case-A grain size distribution of Weingartner & Draine (2001).[2] The model provides the wavelength-dependent extinction cross section, single-scattering albedo ω_λ, and scattering asymmetry parameter $g_\lambda = \langle\cos\xi\rangle$. The angular redistribution of scattered light is described by a Henyey–Greenstein phase function (Henyey & Greenstein 1941).

[1] A bootstrap may yield broader uncertainties because it also samples intrinsic spatial variation within the field. We defer an investigation of the DGL spatial variation to a future work.

[2] The Draine (2003) tabulation reduces the WD01 $R_V = 3.1$ grain abundance per H nucleon by a factor of 0.93. For the two log-normal populations of ultrasmall carbonaceous grains, this changes the carbon abundance from 60 to 55.8 ppm of C/H.

The calculation includes extinction along both the path of the incident radiation and the path from the scattering grain to the observer. The scattered intensity can be written schematically as

$$I_{\lambda,\mathrm{sca}}(b) = I_{\lambda,\mathrm{sca}}\left[\omega_\lambda, \phi_\lambda(\cos\xi), \tau_\lambda, b, J_\lambda(z), n_\mathrm{H}(z)\right], \tag{4}$$

where ϕ_λ is the scattering phase function, ξ is the scattering angle, $J_\lambda(z)$ is the local radiation field, and $n_\mathrm{H}(z)$ is the H nucleon density that sets the dust opacity. The full radiative transfer calculation is described in Appendix C.

To compare the scattered light prediction with the measured DGL SED, we normalize it by the 100 μm emission. We computed the thermal dust emission with the `DustEM` package (Compiègne et al. 2011), using the adopted dust model and illuminating radiation field as inputs. For the dust heating prescription, we adopted the silicate–carbonaceous–PAH framework of Draine & Li (2007, hereafter DL07), with the radiation field strength fixed at $U = 1$ and the PAH mass fraction at $q_\mathrm{PAH} = 4.6\%$. For this prescription, the model gives

$$\langle \nu I_\nu \rangle_{100\mu\mathrm{m}} = 0.52 I_\mathrm{TIR}\,, \tag{5}$$

where I_TIR is the total infrared intensity obtained from the dust emission calculation. Eq. (5) provides a corresponding normalization I_{100}. We discuss the dependence on the adopted dust heating prescription in Sec. 7.1.

For a set of model parameters $\boldsymbol{\theta}$, the prediction is

$$\boldsymbol{m}(\boldsymbol{\theta}) = \frac{\langle \lambda I_{\lambda,\mathrm{sca}}(\boldsymbol{\theta})\rangle}{I_{100}}. \tag{6}$$

The scattered spectrum is integrated through the response curve of each band before comparison with the data. The nominal model for the Perseus field is evaluated at $b = -13°$.

5. Bayesian inference

We use the measured UV-to-NIR DGL SED to infer the wavelength-dependent dust scattering properties. Rather than fitting the parameters independently in each band, we describe smooth departures from the fiducial Milky Way dust model with a small number of parameters. This preserves the broad spectral behavior expected for a physical grain population while allowing the flexibility in data to modify the fiducial model.

5.1. Parameterization of the dust properties

We denote the albedo and scattering asymmetry values in the adopted fiducial dust model by $\omega_0(\lambda)$ and $g_0(\lambda)$. Both quantities are physically bounded. We therefore deform the fiducial model in an unconstrained latent space using the logit transform,

$$\mathcal{T}(p) = \ln\left(\frac{p}{1-p}\right), \qquad \mathcal{T}^{-1}(z) = \frac{1}{1+\exp(-z)}, \tag{7}$$

and define $\lambda' \equiv \ln\left(\lambda/\lambda_\mathrm{piv}\right)$ so that the deformation is described by:

$$\omega(\lambda) = \mathcal{T}^{-1}\left\{\mathcal{T}[\omega_0(\lambda)] + W(\lambda)\left(a_\omega + t_\omega\lambda' + c_\omega\lambda'^2\right)\right\}, \tag{8}$$

$$g(\lambda) = \mathcal{T}^{-1}\left\{\mathcal{T}[g_0(\lambda)] + W(\lambda)\left(a_g + t_g\lambda'\right)\right\}. \tag{9}$$

We adopt $\lambda_\mathrm{piv} = 0.8\,\mu$m near the midpoint of the optical-NIR wavelength coverage. The coefficients a_ω and a_g control the amplitudes of the two curves, while t_ω and t_g describe their spectral tilts. The additional coefficient c_ω allows broad curvature change in the albedo. We use a simpler linear deformation for g, since the recovery tests in Appendix D show that its wavelength dependence is more weakly constrained by the broadband SED.

The inferred parameter vector is thus $\boldsymbol{\theta} = (a_\omega, t_\omega, c_\omega, a_g, t_g)$. A smooth window function $W(\lambda)$ in Eqs. (8) and (9) is used to limit the deformation to the wavelength range covered by the data. It is set to unity through the data wavelength coverage and is tapered smoothly to zero at 0.09 μm and 3 μm. Outside this interval, the model returns to the fiducial model. This avoids extrapolating the inferred changes in ω and g into wavelengths that enter the dust-heating calculation but are not constrained by the observed DGL SED.

5.2. I_{100} normalization treatments

Changing the albedo affects both the scattered light and the energy absorbed and re-emitted by the dust. We therefore consider two treatments of the 100 μm normalization in the inference.

In the *dynamic* treatment, the absorbed stellar power is recalculated for each sampled $\boldsymbol{\theta}$, and I_{100} is updated using the dust-heating prescription in Sec. 4. Increasing ω therefore produces more scattered light while leaving less energy budget for thermal emission. The relation between I_TIR and I_{100} remains fixed to that of the adopted $U = 1$, $q_\mathrm{PAH} = 4.6\%$ DL07 model. In the *fixed* treatment, we instead hold $\widetilde{I_{100}} = 6.7$ MJy/sr equal to the mean 100 μm intensity of the Perseus field. This removes the dependence of inference on the normalization calculation.

The two treatments allow us to assess how strongly the inferred ω_λ and g_λ depend on the adopted dust-heating prescription. The dynamic treatment enforces the coupling between scattering and absorption under the adopted prescription, whereas the fixed treatment conditions the scattering inference on the observed FIR intensity. We return to this question in Sec. 7.1.

5.3. Prior

The logit parameterization keeps ω and g within their adopted physical ranges. The albedo must satisfy $0 < \omega < 1$, while in diffuse ISM forward scattering is predominant over the UV–NIR range (Draine 2003b; Gordon 2004; Ienaka et al. 2013) and therefore the range is set to $0 \le g \le 1$. We used broad uniform priors on the albedo deformation coefficients. The recovery tests in Appendix D, however, show that broadband photometry constrains g_λ less strongly than ω_λ, and the two quantities are degenerate. We therefore regularized the g deformation by applying a prior to $\boldsymbol{\theta}_g$ based on the spread among several Milky Way dust models. The construction of the g prior is detailed in Appendix D.3.

5.4. Likelihood

The data vector contains the DGL correlation measurements from FUV through H_E. We use the jackknife covariance derived in Sec. 3.3 for the likelihood calculation.

For the bands treated as detections, we adopt a multivariate Gaussian likelihood:

$$\ln\mathcal{L}_\mathrm{d} = -\frac{1}{2}\left[(\mathbf{y}_\mathrm{d}-\mathbf{y}_\mathrm{d})^\mathsf{T} C_\mathrm{d}^{-1}(\mathbf{y}_\mathrm{d}-\mathbf{m}_\mathrm{d}) + \ln|C_\mathrm{d}| + N_\mathrm{d}\ln(2\pi)\right], \tag{10}$$

where $\mathbf{y}_\mathrm{d}$ **is the measured DGL SED vector for the detection bands,** $\mathbf{m}_\mathrm{d}$ **is the corresponding model prediction,** C_d is the

corresponding submatrix of the covariance and N_d is the number of detection bands.

We treat the $H_{\rm E}$ measurement as an upper limit on the scattered-light component. Diffuse emission at these wavelengths may contain an additional contribution from very small grains (VSG) not included in the scattering model (Sano & Matsuura 2017). Because $H_{\rm E}$ is correlated with the shorter-wavelength measurements, we retain its measured covariance and evaluate the conditional probability that the scattered-light contribution does not exceed the observed value. The full likelihood is therefore

$$\ln \mathcal{L} = \ln \mathcal{L}_{\rm d} + \ln \Phi\Big(y_{H_{\rm E}} \le y_{H_{\rm E}}^{\rm lim} \mid \mathbf{y}, \boldsymbol{\theta}\Big) \,, \tag{11}$$

in which the conditional probability is evaluated using a standard normal cumulative distribution Φ. In the synthetic recovery tests presented in Appendix D, where the mock SED contains only scattered light, $H_{\rm E}$ is treated as a detection.

5.5. Posterior sampling

For the posterior inference, we first located a high-posterior initial point using optimization and then sampled the posterior with the `emcee` package (Foreman-Mackey et al. 2013). For the results in Sec. 6.4, we used 40 walkers initialized with small Gaussian perturbations around the optimization. Each walker was propagated for 4500 steps, and the first 1200 steps were discarded as burn-in. We estimated the autocorrelation time from the chains and used it to choose the thinning interval. We visually assessed convergence from the trace plot, autocorrelation, and the stability of posterior quantiles.

6. Results

6.1. The UV–optical–NIR DGL SED in Perseus

Figure 3 presents the UV-optical-NIR SED measurement ($\nu I_\nu / I_{100}$) of cirrus/DGL in the Perseus cluster. The measurements were obtained from the ratio of the optical–*Planck* and IRIS–*Planck* slopes, and shown in teal stars (GALEX), blue circles (CFHT), and purple squares (*Euclid*). Together, the GALEX, CFHT, and *Euclid* measurements trace a smooth spectrum that rises from the UV, reaches a broad peak across the optical r, i, and $I_{\rm E}$ bands, and then declines drastically in the NIR bands. The CFHT z measurement has a comparatively larger uncertainty due to the lower data quality. The overall SED shape and amplitude are reproduced reasonably well by the nominal D03/MMP83 radiative transfer model (Sec. 4) at $b = -13°$. In particular, the agreement from the UV through the optical supports a picture in which the diffuse emission in the Perseus field is dominated by Galactic starlight scattered by interstellar dust.

For comparison, Fig. 3 includes a collection of DGL measurements in the literature spanning the UV to NIR (Murthy et al. 2010; Matsuoka et al. 2011; Tsumura et al. 2013; Arai et al. 2015; Sano et al. 2015, 2016b; Onishi et al. 2018; Feder et al. 2025). These measurements were obtained with a range of data types, techniques and environments. [3] We also show the optical DGL continua measured with SDSS, BOSS, **and DESI Y3** blank-sky spectra by Brandt & Draine (2012) Chellew et al. (2022), and Saydjari et al. (2026).

Within uncertainties, our measurements show no clear optical excess from extended red emission (ERE) relative to scattered light model. ERE is generally attributed to photoluminescence from interstellar nanoparticles excited by UV photons, and has been detected in some high-latitude diffuse dust clouds (Szomoru & Guhathakurta 1998; Witt & Vijh 2004). **The absence of detectable ERE toward Perseus does not exclude it along other sightlines, where the UV illumination, scattering geometry, and dust properties could differ.**

Despite the good agreement in the UV and optical, the NIR measurements show a mild departure from the model. In particular, the measured $H_{\rm E}$ intensity lies slightly higher than the model, although the difference remains within the uncertainties. One possible contribution is emission from stochastically heated very small grains (VSGs), including the PAHs. Previous NIR DGL studies find that scattered starlight dominates near 1 μm (Arai et al. 2015; Sano et al. 2015; Onishi et al. 2018), while VSG emission can become increasingly important at longer wavelengths (Sano et al. 2016b). Sano & Matsuura (2017) estimated contributions from VSGs to be less than 10% at 1.25 μm and around 20% at 2.2 μm after including variations in the small grain abundance in their model.

To illustrate the possible contribution, we reduce the measured $J_{\rm E}$ and $H_{\rm E}$ intensities by 10% and 20%, respectively, as shown by the magenta open squares in Fig. 3. This **illustrative adjustment** brings the NIR measurements closer to the scattered light prediction. **It is not a physical correction, because** the NIR dust emission depends on the abundance and size distribution of small grains, as well as the intensity and spectral shape of the local radiation field. Separating dust scattered light from VSG emission will require additional NIR or MIR constraints and modeling efforts, which is beyond the scope of this work.

The figure also highlights the different spectral shapes of Galactic cirrus and the Perseus BCG+ICL. We measure the BCG+ICL SED at 60 kpc and rescale it for comparison. A single stellar population (SSP) model with age of 13 Gyr and solar metallicity is overlaid for illustrative purpose, which qualitatively matches the BCG+ICL measurements. The emission remains strong into the NIR, whereas cirrus peaks in the optical and declines drastically in NIR. This contrast provides useful diagnostics for distinguishing the two diffuse components. We discuss the comparison using colors in Sec. 6.3.

6.2. From dust energy budget to dust scattered light

Figure 4 shows the measured and modeled scattered-light SED together with the thermal dust emission on a common frequency axis, so that absorbed and re-emitted energy can be compared. The modified blackbody (MBB) curve provides a simple description of equilibrium thermal emission from large grains, while the DL07 curve includes emission produced by PAHs and other small grains calculated for $U = 0.7$ and $q_{\rm PAH} = 4.6\%$.[4] The dust emission measurements from *Planck* HFI prod-

[3] The Murthy et al. (2010) points in UV are based on archival GALEX imaging. In the NIR, the Arai et al. (2015) points are based on CIBER low-resolution spectrometer at high latitude, whereas the Onishi et al. (2018) points are based on MIRIS imaging of high-latitude clouds. The measurements of Sano et al. (2015, 2016b) are based on all-sky COBE/DIRBE 1.25 to 4.9 μm data (Sano et al. 2015, 2016b) at high latitude. Tsumura et al. (2013) measurements are based on low-resolution spectroscopy with AKARI/IRC. Two NIR DGL constraints are given by Feder et al. (2025) using the cross-power spectra between the CIBER or the Spitzer/IRAC maps and the CIB-corrected dust map (CSFD).

[4] **We found $U = 0.7$ better match the Planck HFI data. Note that we adopt $U = 1$ for the nominal model because DL07 define U relative to the MMP83 ISRF, which also sets the midplane illumination in our model. This U difference affects the dynamic-I_{100} by $\sim 10\%$,**

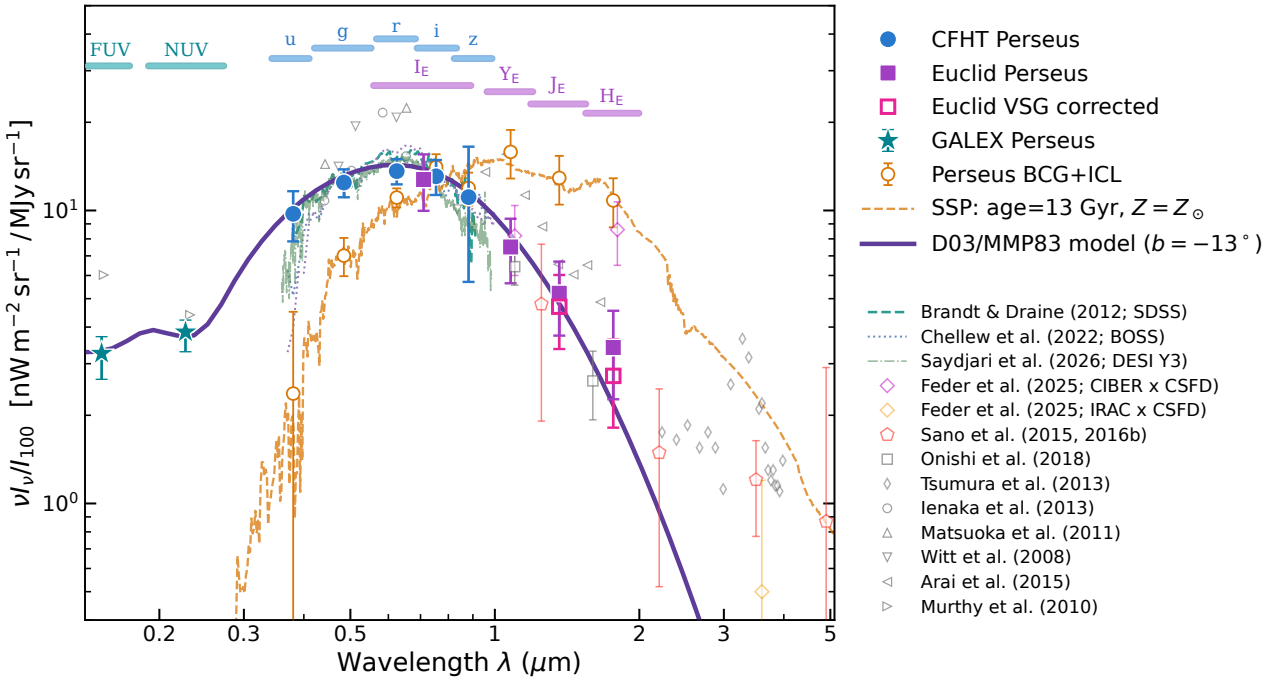


Fig. 3: The cirrus/DGL SED in the Perseus field normalized by FIR dust tracer, expressed as $\nu I_\nu / I_{100}$. The measurements include GALEX FUV and NUV, CFHT *ugriz*, and *Euclid* VIS and NISP bands. Gray symbols show literature measurements of DGL or diffuse dust clouds. The thick purple curve is the nominal D03/MMP83 radiative transfer model at $b = -13°$. Magenta open squares show the illustrative VSG correction, in which the measured J_E and H_E intensities are reduced by 10% and 20%, respectively. The orange circles show the Perseus BCG+ICL measurements at 60 kpc, while the orange dashed curve shows a 13-Gyr SSP spectrum with solar metallicity. The BCG+ICL measurements and SSP model are rescaled to compare their spectral shapes with the DGL. The teal dashed curve shows the rescaled BD12 DGL continuum using SDSS. The blue dotted and dash-dotted curves show the rescaled northern and southern continua obtained by Chellew et al. (2022) using BOSS and that obtained by Saydjari et al. (2026) using DESI Y3 data. **The continua from SDSS, BOSS, and DESI sky spectra are rescaled to match CFHT-*g*.**

ucts match well with a thermal MBB model of $T = 18\,K$ and $\beta = 1.8$.

For comparison, we show the input MMP83 ISRF together with the ISRF weighted by the D03 scattering and absorption cross-sections, C_{sca} and C_{abs}. Their normalization is set by energy balance: the frequency-integrated absorbed power, represented by the area under the orange dotted curve (ISRF×C_{abs}), is matched to the integrated thermal dust emission, which is the area under the *Planck* thermal curve. The ISRF and ISRF×C_{sca} curves were scaled by the same factor.

The ISRF×C_{sca} spectrum provides a useful reference for the wavelength dependence of dust scattering. Its difference from the radiative transfer model is most prominent in the UV and optical, where attenuation, scattering geometry, and the anisotropy of the phase function have a stronger effect on the light reaching the observer. Toward the NIR where attenuation is weak, the two curves become more similar. This comparison shows how the observed DGL SED is shaped not only by the dust scattering cross-section, but also by the geometry and radiative transfer through the Galactic dust layer.

6.3. Optical/NIR colors of cirrus in Perseus

For extragalactic analyses of low surface brightness features, measurements are usually expressed in mag/arcsec^2, and their colors are of greater interest. Figure 5 shows several optical and NIR color combinations derived from the Perseus DGL SED **using the bandpass-averaged intensity** I_ν. These colors are independent of the FIR normalization and directly depict the SED shape.

The measured optical/NIR colors are shown as purple squares. The VSG-corrected colors are marked as magenta open squares. The color derived from the fiducial model is shown as the purple diamond. We also show the colors predicted by WD01 dust models with $R_V = 3.1$, 4.0, and 5.5 over a range of latitudes from $|b| = 15°$ to 75°. The models have similar optical colors, and become redder in NIR colors from low to high Galactic latitudes. Red arrows indicate how an additional ERE component could redden the colors. *Euclid* NIR colors are not affected by ERE reddening.

For comparison, we show the Perseus BCG+ICL colors between 20 and 100 kpc, using the *Euclid* measurements of Kluge et al. (2025) and the CFHT colors derived from the transferred models. We corrected these measurements for Galactic extinction using the SFD98 dust map (Schlegel et al. 1998). The BCG+ICL becomes slightly redder with radius in $g - r$, $r - i$, and $I_E - Y_E$, while becoming bluer in $Y_E - J_E$ and $J_E - H_E$. We also show stellar population synthesis (SPS) tracks with an exponentially declining star formation history, **an e-folding time of** 1 Gyr, spanning ages of 0.3–13 Gyr and metallicities from $Z/Z_\odot = 0.03$ to 2. The SPS models are generated using the `fsps` package (Conroy & Gunn 2010).

In the CFHT $g - r$ versus $r - i$ plane, the Perseus cirrus lies near the region occupied by intermediate-age, metal-poor stellar

but do not affect the fixed-I_{100} treatment. Future work will take into account the possible variation in U.

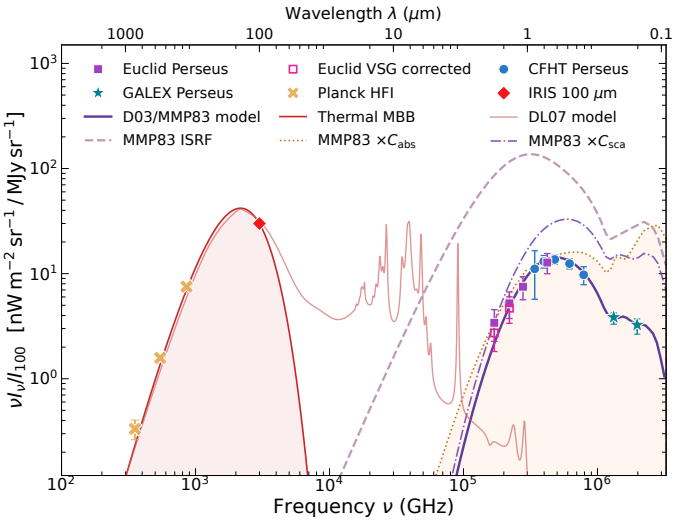


Fig. 4: The Perseus DGL SED as a function of frequency. The measurements and the radiative transfer curve are the same as in Fig. 3. The light red curve shows the DL07 dust emission model spectrum with $U = 0.7$ and $q_{\rm PAH} = 4.6\%$, and the dark red curve shows the thermal-only modified blackbody emission with $T = 18\,K$ and $\beta = 1.8$, normalized by the predicted I_{100}. The orange markers show the emission measured from *Planck* HFI maps. The purple dashed curve shows the input MMP83 ISRF. The purple dash-dotted and orange dotted curves are the same radiation field multiplied by the D03 scattering and absorption cross-sections, respectively. An energy-balance normalization is applied such that the absorption (orange area) equals the corresponding integral of the thermal emission (red area).

populations. The Stripe 82 cirrus measurements of Román et al. (2020) are somewhat **redder** in $g-r$, while their $r-i$ colors are similar. Differences in illumination, dust properties, scattering geometry, and a possible ERE contribution could all cause the offset between the two sightlines.

In the $I_{\rm E}-Y_{\rm E}$ versus $Y_{\rm E}-J_{\rm E}$, the measured cirrus agrees with the model in $I_{\rm E}-Y_{\rm E}$, while being consistent in $Y_{\rm E}-J_{\rm E}$ only after the VSG correction. Its $I_{\rm E}-Y_{\rm E}$ color is bluer than most of the SPS models. In the NISP-only $Y_{\rm E}-J_{\rm E}$ vs $J_{\rm E}-H_{\rm E}$ plane, the measured cirrus is redder than the nominal dust model but remains bluer than most of the SPS tracks.

Finally, when combining *Euclid* $I_{\rm E}-J_{\rm E}$ and CFHT $g-r$ colors, the cirrus measurement is well separated from the SPS models. The measurement is also close to the prediction. The dashed line in the lower right panel illustrates this separation. A possible ERE contribution would move the cirrus further toward the lower-right direction in this diagram rather than toward the bulk of the SPS models.

Despite the relatively large uncertainties in the NIR bands, in all color-color diagrams the BCG+ICL colors occupy a distinct region from both the cirrus measurements and models. This suggests that one can use optical and NIR colors to help distinguish Galactic cirrus from diffuse stellar emission. This idea follows the experiment presented in Román et al. (2020) using $g-r$ vs $r-i$, while the separation becomes clearer when *Euclid* NIR data are combined with optical measurements.

6.4. Constraints on dust properties

We now use the Perseus DGL SED measurements presented in Sec. 6.1 to constrain the wavelength dependence of the dust scattering properties. Figure 6 summarizes the posterior inference for the fixed and dynamic I_{100} treatments introduced in Sec. 5.2. In Fig. 6 panel (a) and (b), the colored curves show the posterior medians of ω_λ and g_λ, with shaded areas indicating the 16th–84th percentiles. The fiducial D03/WD01 $R_V = 3.1$ model is shown by the black dashed curve. The return of the inferred curves to the fiducial model at the wavelength boundaries is by construction. Panel (c) shows the corresponding posterior DGL SED and panel (d) shows the residual normalized by the uncertainty $\sigma_{\rm meas}$. Both treatments reproduce the overall UV-optical-NIR shape of the observed SED, where $H_{\rm E}$ is treated as a censored measurement **due to the excess likely from non-scattering emission**.

The albedo is the dust parameter best constrained by the broadband SED. The values evaluated at the photometric bands are listed in Table 1. In the FUV, both normalization treatments favor an albedo $\simeq$ 0.42–0.47, which is above the D03/WD01 curve. Around the NUV the result is closer to the fiducial model. Across the optical bands, the fixed-I_{100} solution remains below the D03/WD01 model with $\omega \simeq 0.56$–0.58, whereas the dynamic-I_{100} solution gives $\omega \simeq 0.63$–0.65 and lies much closer to the fiducial curve. In the NIR, the constraints become broader, as the signal weakens and the photometric uncertainties increase, with the inferred values around $\omega \simeq 0.5$–0.6. Note that $H_{\rm E}$ provides only an upper limit on the scattering component because additional dust emission may contribute at these wavelengths.

The difference between the fixed and dynamic solutions illustrates the sensitivity of the inferred albedo to the adopted dust heating prescription. The dynamic-I_{100} treatment, in which increasing ω also reduces the absorbed energy, favors a higher albedo than the fixed-I_{100} treatment. We discuss this dependence further in Sec. 7.1.

The scattering asymmetry g_λ behaves differently. The fixed and dynamic treatments give nearly identical posterior curves, both broadly consistent with the fiducial D03 model. Their widths, however, are close to the model-envelope prior (dotted curve) introduced in Appendix D.3. The Perseus broadband SED therefore adds relatively little information on g, so the values in Table 1 **reflect the spread of the reference models** rather than direct constraints on g_λ.

This behavior is consistent with the recovery tests in Appendix D, which show that changes in g can be partly compensated by slight changes in ω and still produce very similar broadband SEDs. Measurements over a single sightline therefore do not strongly constrain the scattering phase function. Previous DGL studies reached a similar conclusion: strong constraints on g generally come from measurements across different Galactic environments (Mattila et al. 2018, 2023).

A recent low-latitude example is given by Sano et al. (2025), who found that optical DGL near the Galactic center decreases with increasing optical depth. They could reproduce this trend using a dust slab radiative transfer model with a broad range of ω and g values once the background illumination was varied. Although the Perseus sightline is less optically thick, its relatively low Galactic latitude likewise makes the illumination geometry more complex than in classical high-latitude DGL fields. This reinforces the difficulty of separating dust scattering properties from the geometry and intensity of the radiation field using a single sightline.

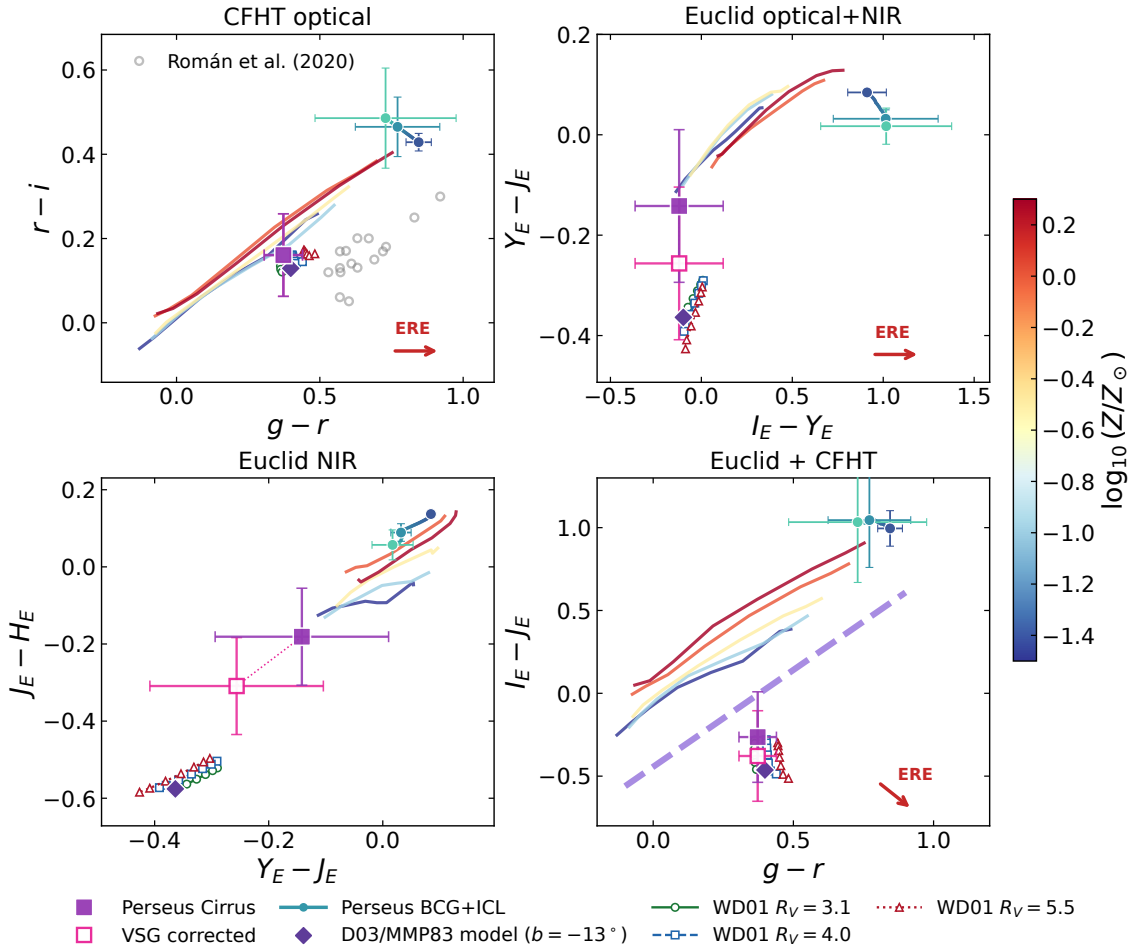


Fig. 5: Color-color diagram of Galactic cirrus in the Perseus field, compared with stellar population models and BCG+ICL colors. Filled purple squares show the measured cirrus colors, while open magenta squares show the illustrative VSG correction obtained by reducing the measured J_E and H_E intensities. The purple diamond is the fiducial D03/MMP83 prediction at $b = -13°$. The green, blue, and red loci show the corresponding WD01 predictions for $R_V = 3.1$, 4.0, and 5.5 over $|b| = 15$–$75°$, illustrating the effect of varying scattering geometry and dust properties. The BCG+ICL color loci extend from 20 to 100 kpc, with markers shown at 20/60/100 kpc. The colored stellar population tracks show exponentially declining models with $\tau = 1$ Gyr, ages of 0.3–13 Gyr, and metallicities from $Z/Z_\odot \simeq 0.03$ to 2. Gray circles in the CFHT panel show the cirrus measurements from Román et al. (2020) using SDSS Stripe82 dataset. The dashed purple line in the *Euclid*+CFHT panel is a visual separation parallel to the stellar population tracks between SPS models and cirrus measurements / models. Red arrows indicate the direction of a possible additional ERE contribution in r, i, and I_E. The NISP colors are not affected by ERE reddening.

Overall, the Perseus broadband SED constrains ω_λ more directly than g_λ. Stronger constraints on g_λ will likely require measurements with substantially improved precision and across multiple sightlines spanning different Galactic latitudes and illumination geometries.

6.5. Comparison with literature dust albedo measurements

Since the Perseus SED constrains ω_λ more directly than g_λ, we compare the inferred albedo with previous measurements in the literature and several Milky Way dust models in Fig. 7. The literature compilation covers diffuse high-latitude cirrus, translucent clouds, and a small number of denser regions. Direct albedo measurements in the NIR are particularly limited.

In the UV, measurements span a range of Galactic environments, from high-latitude diffuse fields such as Draco, to regions closer to the Galactic plane (Sujatha et al. 2010; Lim et al. 2013; Akshaya et al. 2019; Murthy et al. 2020). Lim et al. (2013) modeled FUV scattering across the Taurus–Auriga–Perseus complex and obtained $\omega = 0.42 \pm 0.05$, which is in good agreement with our FUV result. Their analysis covered the **ERO Perseus field**, and also showed that the inferred scattering properties depend strongly on the relative geometry of the illuminating stars and dust.

We include more recent optical measurements, which show a broad range. Mattila et al. (2018) found $\omega \sim 0.6$–0.7 using observations of translucent clouds LDN 1780, 1642 and LBN 406. Zhao et al. (2024) obtained lower values, typically $\omega \sim 0.4$–0.5 in g and r, using DESI Legacy Survey imaging data at $|b| \geq 30°$, albeit with large field-to-field scatter. Togi et al. (2017) derived upper limits with observations of the denser cloud LDN 1489.

In the NIR, direct constraints remain sparse. Lehtinen & Mattila (1996) inferred high J- and H-band albedos for the Thumbprint Nebula, although its dense environment may host grain populations different from those in diffuse cirrus.

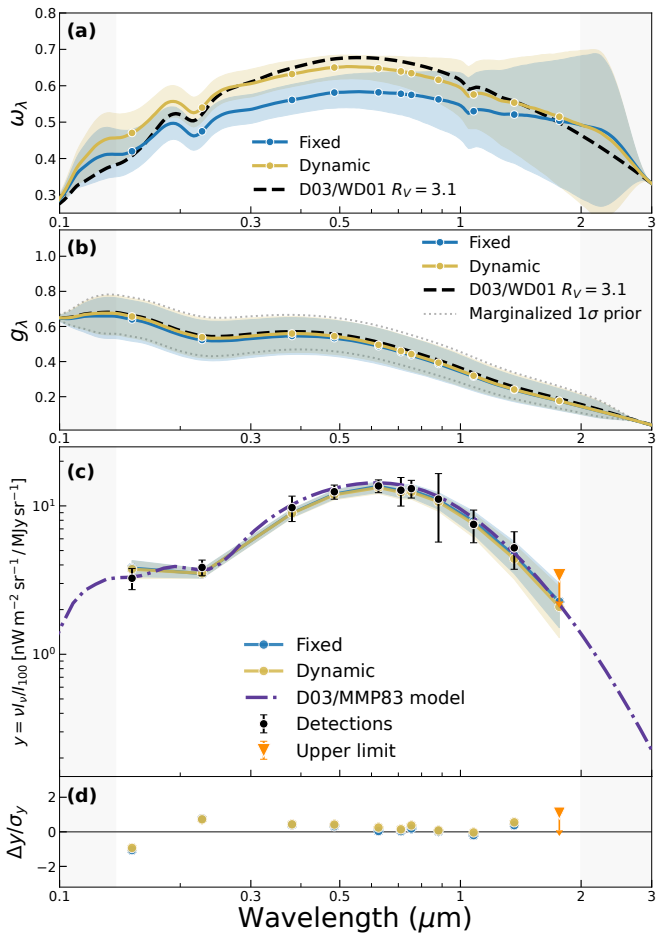


Fig. 6: Posterior constraints on the dust scattering properties and DGL SED in Perseus. **(a)**: single-scattering albedo ω_λ. **(b)**: scattering asymmetry parameter g_λ. Blue and yellow curves show the posterior medians for the fixed and dynamic I_{100} treatments, respectively, with shaded regions showing the 16%–84% quantile ranges. The dashed black curves show the fiducial D03/WD01 $R_V = 3.1$ model. The dotted curve in (b) shows the 16%–84% range of the marginalized g prior. **(c)**: corresponding posterior DGL SED compared with the measurement. Black points show detections and the orange marker denotes the censored H_E measurement. **(d)**: Residuals of measurement and posterior median SED, normalized by the detection uncertainty σ_y. The gray areas indicate the wavelengths outside the data.

The theoretical curves span a comparable range. The D03/WD01 models vary with R_V through changes in the adopted grain-size distribution (Weingartner & Draine 2001; Draine 2003b), with higher-R_V models generally predicting larger optical and NIR albedos. The Zubko et al. (2004, hereafter ZDA04) BARE–GR–S model adopts a graphite–silicate–PAH mixture. The DustEM model shown here refers to the diffuse high latitude dust model in Compiègne et al. (2011), which includes PAHs, small and large amorphous carbon (`SamC` and `LamC`), and amorphous silicates (`aSil`). The Astrodust model instead describes the traditional separate large carbonaceous and silicate populations with a composite material together with PAHs (Hensley & Draine 2023). Differences in grain composition and size distribution lead to appreciable differences in the predicted albedo, particularly in the NIR.

Table 1: Posterior median dust-scattering parameters and 68% credible intervals, defined by the 16th and 84th percentiles, evaluated at the UV–NIR photometric bands for the fixed and dynamic I_{100} normalization treatments.

	ω_λ		g_λ	
Band	Fixed	Dynamic	Fixed	Dynamic
FUV	$0.42^{+0.06}_{-0.06}$	$0.47^{+0.06}_{-0.06}$	$0.64^{+0.11}_{-0.13}$	$0.65^{+0.10}_{-0.13}$
NUV	$0.48^{+0.04}_{-0.04}$	$0.54^{+0.04}_{-0.04}$	$0.52^{+0.11}_{-0.12}$	$0.54^{+0.11}_{-0.12}$
u	$0.56^{+0.04}_{-0.04}$	$0.63^{+0.04}_{-0.04}$	$0.55^{+0.10}_{-0.11}$	$0.56^{+0.10}_{-0.11}$
g	$0.58^{+0.05}_{-0.04}$	$0.65^{+0.03}_{-0.03}$	$0.53^{+0.09}_{-0.10}$	$0.54^{+0.09}_{-0.10}$
r	$0.58^{+0.06}_{-0.05}$	$0.65^{+0.03}_{-0.04}$	$0.49^{+0.09}_{-0.10}$	$0.50^{+0.09}_{-0.10}$
i	$0.58^{+0.07}_{-0.06}$	$0.64^{+0.05}_{-0.05}$	$0.44^{+0.09}_{-0.10}$	$0.44^{+0.09}_{-0.10}$
z	$0.56^{+0.08}_{-0.08}$	$0.62^{+0.06}_{-0.06}$	$0.39^{+0.09}_{-0.09}$	$0.40^{+0.09}_{-0.09}$
I_E	$0.58^{+0.06}_{-0.06}$	$0.64^{+0.04}_{-0.04}$	$0.45^{+0.09}_{-0.10}$	$0.46^{+0.09}_{-0.10}$
Y_E	$0.53^{+0.10}_{-0.10}$	$0.58^{+0.09}_{-0.10}$	$0.32^{+0.09}_{-0.08}$	$0.32^{+0.09}_{-0.08}$
J_E	$0.52^{+0.13}_{-0.13}$	$0.56^{+0.12}_{-0.14}$	$0.24^{+0.08}_{-0.07}$	$0.25^{+0.08}_{-0.07}$
H_E	$0.50^{+0.17}_{-0.18}$	$0.52^{+0.18}_{-0.19}$	$0.18^{+0.07}_{-0.05}$	$0.18^{+0.07}_{-0.06}$

In the optical, the dynamic-I_{100} solution in Perseus lies close to the D03/WD01 $R_V = 3.1$ curve, with $\omega \simeq 0.63$–0.65 from u through i. The fixed-I_{100} solution is lower, with $\omega \simeq 0.56$–0.58, and is closer to the DustEM model. The difference between the two inference treatments, however, is comparable to the spread among dust models.

In the NIR, our *Euclid* Y_E, J_E, and H_E albedos are consistent with the D03/WD01 models, the DustEM model, and the Astrodust model within the uncertainties. They lie below the measurements of Lehtinen & Mattila (1996), which were obtained in a denser dust environment. Wider-area measurements with *Euclid* and *Roman* will provide opportunities for better constraining the NIR dust albedos, investigating the albedo spatial variations, and disentangling the coupling between dust properties and illumination or scattering geometry.

7. Discussion

7.1. Dust scattering and extinction in the Perseus field

7.1.1. Dust scattering and model dependence

The broad optical peak and decline toward NIR wavelengths in the Perseus DGL SED shown in Fig. 3 support dust scattered starlight as the main origin of the optical DGL, as established by previous photometric and spectroscopic studies (e.g., (Witt et al. 2008; Ienaka et al. 2013; Brandt & Draine 2012; Chellew et al. 2022)). *Euclid* extends this picture by resolving the cirrus at high angular resolution from the optical into the NIR with its excellent systematic control in diffuse emission. **Its NISP coverage complements earlier NIR measurements from infrared satellites and rocket experiments** (Arai et al. 2015; Sano et al. 2015, 2016b; Onishi et al. 2018), and provides a resolved view of DGL in the wavelength range where cirrus is less explored. This is useful both for constraining the grain population and for distinguishing cirrus from diffuse extragalactic light.

Our inference shows that the derived albedo depends on the dust heating prescription, that is, how the FIR normalization is modeled. Across the optical bands, the inferred ω changes from approximately 0.56–0.58 for the fixed-I_{100} treatment to

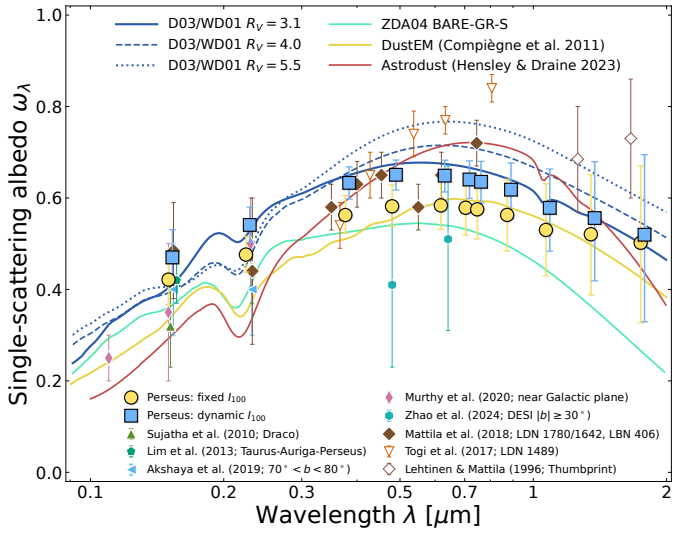


Fig. 7: Single-scattering albedo inferred for the Perseus field compared with literature measurements and Milky Way dust models. **Yellow circles and blue squares** show the fixed and dynamic I_{100} Perseus inference, respectively. The literature measurements cover diffuse ISM, translucent clouds, and dense globules, as indicated in the legend. Measurements from Togi et al. (2017) denote upper limits. Model curves show D03/WD01 model for R_V = 3.1, 4.0, and 5.5, ZDA04 BARE–GR–S, DustEM, and astrodust. Note measurements from different studies adopt different radiation fields, geometries, and assumptions about the phase function.

0.63–0.65 when I_{100} is computed dynamically. Changing ω affects both the scattered intensity and the energy budget available for thermal emission. Sano & Matsuura (2017) showed that the prescription of 100 μm emission has a non-negligible effect on the latitude dependence of NIR DGL given its low intensity and contribution from non-scattering components. Our dynamic normalization adopts a simplified heating prescription, which does not recalculate the full thermal spectrum as the illumination or grain populations change. The inferred scattering parameters therefore should be treated as conditional on this approximation.

Illumination introduces a related model uncertainty. Brandt & Draine (2012) compared results of the MMP83 spectrum with that of a stellar population spectrum from Bruzual & Charlot (2003). The greater UV output in the BC03 model leads to increased dust heating and a reduced optical-to-FIR ratio. More recently, Bianchi (2024) reconstructed a local ISRF using Gaia DR3 and found it redder than MMP83 with 30% more integrated UV-to-NIR energy. These results demonstrate that the choice of reference illumination affects model comparison and inference.

The anisotropy of the scattering matters separately from the illumination. The classic Henyey–Greenstein phase function has been found to be a reasonable approximation at 0.4–1 μm, but with a poor description in the UV and NIR (Draine 2003b; Baes et al. 2022). **Broadband measurements alone** place weak constraints on the scattering asymmetry and the phase function. This could matter at the relatively low Galactic latitude of Perseus, where dust scattering takes place in a complicated illumination geometry that receives illumination from different directions, similar to the environment studied by Sano et al. (2025). Variations in illumination and optical depth can likely explain the variations in the optical-to-FIR ratios (Ienaka et al. 2013; Mattila et al. 2023).

Additional dust emission could also contribute to cirrus SED and color. One possible source is the photoluminescence excited by UV photons, known as the ERE. Although its carrier remains uncertain, ERE has been detected in reflection nebulae and in the diffuse ISM (Szomoru & Guhathakurta 1998; Witt et al. 2008; Witt & Lai 2020), **producing redder optical colors**. Our broadband photometry does not confirm an ERE component in the Perseus field, although its strength may vary among sightlines (Brandt & Draine 2012; Chellew et al. 2022). The mild NIR excess relative to the scattering model may indicate an emission contribution from stochastically heated small grains, making the optical-NIR color redder. Both effects are illustrated in Fig. 5. In the UV, molecular fluorescence can also produce excess in diffuse FUV emission (Mattila et al. 2023; Burkhart et al. 2025).

7.1.2. Dust scattered light as a probe of diffuse ISM column density

We next examine the extent to which dust scattered light can probe diffuse ISM column density. Our model assumes an approximately linear FIR correlation and single-scattering. In the optically thin limit, with fixed illumination and grain properties, scattered intensity increases approximately in proportion to column density. Where attenuation becomes prominent, multiple scattering becomes non-negligible and the surface brightness can flatten and then decline as column density increases (Mattila et al. 2023), and deviate from a linear relation with FIR emission (Zhang et al. 2023). A different radiative transfer model is thus preferred for the optically thick regime (Sano et al. 2025). Perseus lies at a relatively low Galactic latitude ($b = -13°$), posing a challenge to our assumptions, especially at bluer wavelengths.

We therefore checked the validity of linear correlation with dust reddening. Over the reddening range $E(B-V)_{\rm Planck} \simeq 0.13$–$0.20$ in the Perseus field, which corresponds to $A_V \simeq 0.39$–0.63 for R_V = 3.1, the optical/NIR intensity increases linearly and shows no clear turnover, following similar tests in Liu et al. (2025a).

Figure 8 illustrates the ISM column density regime that can be probed through dust scattered light. We convert the nominal 1σ surface brightness sensitivities[5] of GALEX, CFHT, and *Euclid* into equivalent hydrogen column densities ($N_{\rm H}$) using the measured Perseus DGL correlations. For $J_{\rm E}$ and $H_{\rm E}$, we use the measured DGL intensities without VSG correction. We adopt the high-latitude H I calibration of Boulanger et al. (1996): $N_{\rm H}/(10^{20}\,{\rm cm^{-2}}) = 1.89\times I_{100}\,/({\rm MJy/sr})$ The resulting Euclid sensitivities extend well into the diffuse ISM regime and below the characteristic foreground level toward Perseus. These values should be treated as illustrative equivalent sensitivities, since the conversion depends on illumination, dust emissivity, and dust temperature, and the quoted depths use different conventions.

The resolved cirrus maps also reveal dust structures on scales finer than standard reddening maps, motivating future attempts to better characterize dust extinction in diffuse ISM. Converting scattered light intensity directly into reddening, however, re-

[5] The adopted CFHT and Euclid surface-brightness limits are on $10'' \times 10''$ scales following the published values in Euclid Collaboration: Scaramella et al. (2022) and Cuillandre et al. (2025b); the *Euclid* Deep Field (EDF) values are forecast depths. For GALEX, we adopt an exposure typical of the Nearby Galaxy Survey ($\sim$ 1.7 ks) and scale the surface brightness limits reported by Goddard et al. (2010) to $10'' \times 10''$ scale. Note that advanced processing of GALEX imaging can reach substantially fainter limits (Ruiz Cejudo et al. 2025).

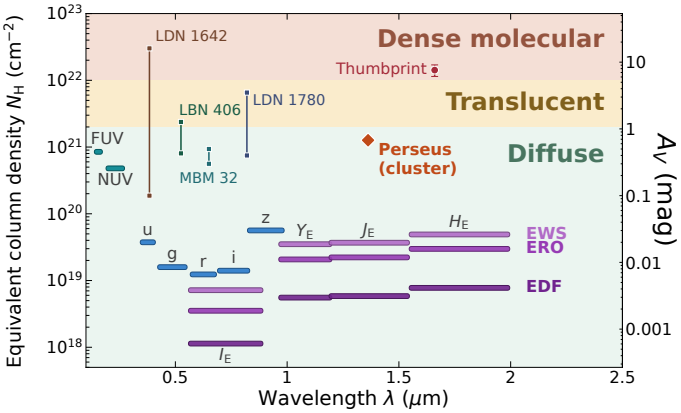


Fig. 8: Equivalent 1σ hydrogen column density sensitivities estimated from GALEX, CFHT (deep), and Euclid (ERO/EWS/EDF) surface brightness limits and the measured Perseus DGL. *Euclid* can probe well into the diffuse ISM regime. Horizontal sticks show the filter bandpasses. Euclid EWS, ERO, and forecast EDF limits are shown in light to dark purple. Reference markers indicate extinction ranges of representative diffuse/translucent clouds, the core of the dense cloud Thumbprint, and the mean foreground level toward the Perseus cluster; their wavelength positions are schematic. For *Euclid*, the cirrus sensitivity is lower in the NISP bands than in VIS because of the shallower surface brightness limits and the intrinsically fainter NIR dust scattered signal.

quires calibration against dust column density and independent control of the illumination and grain properties. Such analysis would benefit from reddening measurements of background stars using *Euclid* data.

7.2. Systematics in processing and measurement

The correlation method in Sec. 3.2 is less sensitive to absolute sky offsets than a direct intensity measurement. Residual structures that happen to spatially correlate with the dust, however, can still bias the measurement. The main concerns are therefore large-scale sky patterns, imperfect separation of cluster light and cirrus, and residual source emission through masking and reconstruction.

The large-scale sky pattern includes zodiacal light, atmospheric emission in the CFHT data, flat-field residuals, and instrumental stray light. After subtracting the `ZodiSURF` model, any remaining zodiacal-light residual is expected to vary smoothly across the $0.7 \times 0.7\,\mathrm{deg}^2$ field-of-view. Airglow, flat-field residuals and instrumental stray light behave in a similar way. We account for such a pattern with the low-order sky term in Eq. (2). This comes with a trade-off, because cirrus itself contains power on large angular scales. As a check, we repeated the measurement after subtracting a simple plane from the images and fitting the dust correlation without an additional sky term, and obtained similar results. Therefore, the measured correlation slopes are not driven by the exact treatment of the large-scale sky pattern. The absolute cirrus zero-level is less well constrained. Setting it by extrapolating the correlation to zero dust radiance is a calibration assumption (Liu et al. 2023, 2025a). Our correlation measurement is much less sensitive to this choice. Differential measurements against dust-poor reference regions provide an alternative approach (Mattila et al. 2018; Sano et al. 2025).

The subtraction of the BCG and ICL is a more important, though field-specific, systematic for Perseus. Instead of modeling this light independently in the CFHT data, we transferred the *Euclid* I_E BCG+ICL model with a radial color correction (Appendix A.4). This assumes that the broad BCG+ICL morphology is well represented by the VIS template and that any remaining color gradient outside the fitted region is small. Diffuse cluster substructures may remain, while cirrus absorbed into the ICL model may lead to some oversubtraction. We varied the exclusion region around the BCG, the BCG–ICL transition radius, and the radial range used in the scaling fit and found no significant change in the DGL SED. The color comparison between cirrus and cluster light is more sensitive to such residuals; slight changes in the SED shape could lead to a larger shift in color.

Source masking and infilling also introduce systematics, especially in a crowded field like Perseus. `DeNeb` and the subsequent morphological filtering and halo subtraction remove most compact emission and localized residuals while retaining diffuse cirrus, but some low surface brightness structure inevitably remains. *Euclid* has relatively well controlled extended PSF wings (Cuillandre et al. 2025b), whereas the CFHT images show more prominent stellar ghosts and PSF wings, especially in *u*. At redder wavelengths, extended galaxy halos and intragroup light become more important. The processing reduces these features, although their faint outskirts remain difficult to separate where they overlap the cirrus. Because such residuals are not associated with the dust, to first order they increase the scatter of the correlation rather than produce a coherent DGL signal. The jackknife test shows no evidence that the measured SED is dominated by a small number of localized structures.

The nfilled regions in Fig. 2 are only plausible reconstructions and may propagate artifacts that were produced by imperfect masking. For the DGL measurements, infilled pixels were excluded from the correlation and were used mainly to support operations such as convolution. Their direct effect on the measured correlation is therefore limited, although some artifacts may still propagate through beam matching. The effect could be larger for structural measurements such as the power spectrum, which is deferred for future investigation.

Finally, the FIR dust maps contain separate systematics. We use *Planck* radiance for the optical/NIR correlation and a Planck–IRIS correlation to convert normalization to I_{100}. Spatial changes in dust temperature or emissivity lead to the scatter in the conversion between radiance and 100 μm intensity. Residual CIB structure and calibration differences can also add uncertainties, although the bias has been reduced compared to using a direct correlation between the optical/NIR and I_{100} or using intensity ratios.

7.3. Galactic cirrus as a foreground for low surface brightness science

The color diagrams in Fig. 5 suggest that cirrus and diffuse stellar emission can be well distinguished by combining optical photometry with NIR detections. Optical colors alone provide some separation, but cirrus can overlap with metal-poor, intermediate age stellar populations. The NISP-only colors have relatively large uncertainties and may be affected by small-grain emission at longer wavelengths. The cleanest separation is found where CFHT $g - r$ is combined with *Euclid* $I_E - J_E$ color (**Fig. 5**). Despite the broad bandpass of VIS, the measured cirrus and the DGL models occupy distinct regions from the SPS models and the BCG+ICL sequence.

Previous studies have explored similar distinctions using optical colors. Román et al. (2020) found cirrus in Stripe 82 to be relatively blue in $r - i$ compared to SDSS galaxies at fixed $g - r$ with $g - r \simeq 0.5$–0.9 and $r - i \simeq 0.05$–0.3, while measurements around the HCG 86 group by Ragusa et al. (2021) found a redder $r - i \simeq 0.4$–0.9 at similar $g - r \simeq 0.5$–0.8. Liu et al. (2025a) measured $g - r \simeq 0.6$–0.7 for cirrus in a $\sim$10 deg^2 imaging dataset obtained by the Dragonfly telescope, about 0.1–0.15 mag redder than typical faint dwarf galaxies,[6] and about 0.3 mag redder than the ultra-faint dwarf satellite And XXII. These measurements show that optical color helps separate stellar emission from cirrus, but does not fully remove the ambiguity.

Part of this variation is physical. Optical cirrus colors depend on illumination, optical depth, dust properties, and possible ERE. More recently, models of turbulent dust clouds show that increasing optical depth generally yields redder optical colors, while higher density fluctuations can shift the mean color in the opposite direction (Seon et al. 2025). ERE does not affect the NISP bands, although small-grain emission becomes a concern in the NIR. At wavelengths around 1 μm, this contribution is expected to remain modest (<10%) relative to the scattered light component (Sano & Matsuura 2017). The boundary drawn in Fig. 5 should therefore be regarded as illustrative rather than as a universal cirrus classifier.

The Perseus and Abell 2390 ERO fields also demonstrate why low-resolution infrared templates alone cannot fully remove cirrus from deep optical imaging. In Perseus, Kluge et al. (2025) used a scaled WISE 12 μm map to mitigate large-scale cirrus contamination in their analysis. In Abell 2390, direct subtraction of the WISE template left visible small-scale residuals, because cirrus structures do not match exactly between the MIR and optical/NIR. Ellien et al. (2025) instead used the multiscale wavelet method DAWIS to model the diffuse emission and extract ICL. These examples show that infrared maps are helpful for *locating* Galactic cirrus, but are not always adequate by themselves to recover its structure at the *Euclid* resolution.

This picture is consistent with our earlier morphological analysis (Liu et al. 2025b), where optical cirrus structures are found to most closely follow FIR dust emission across spatial scales, while differences became more apparent for finer structures. Pixel-by-pixel subtraction of FIR templates is limited by their coarse resolution and by contaminants such as the CIB. We also showed that filamentary morphology and higher-order image statistics can help identify cirrus from more localized extragalactic structures such as tidal features. The present color measurement is complementary to such information. In practice, a useful separation can combine three observables: FIR emission, which locates dust on large scales; resolved morphology, which traces fine structures; and optical–NIR colors, which tells the spectral difference of cirrus and stellar populations. Our future work will aim to combine these pieces of information.

The problem will become more common as deep surveys cover larger areas of sky. Although the EWS avoids the dustiest regions, *Euclid* will nonetheless encounter cirrus over parts of its area, affecting measurements of ICL, stellar halos, tidal debris, and faint galaxies. *Roman* and CSST will face similar foregrounds in deep optical and NIR imaging, with Rubin providing complementary ground-based data. SPHEREx, with its all-sky spectroscopy in the NIR, is able to sample the transition from scattering to small-grain emission. Together, these datasets should make it possible to characterize how the cirrus SED changes across Galactic environments and to build better foreground models for future low surface brightness studies.

[6] The reference dwarf galaxy model has $\log(M_\star/M_\odot) = 8$, [Fe/H] = −1.5, and an age of 9 Gyr. A parameter grid was also explored. See details in Liu et al. (2025a).

8. Conclusions

We have measured and modeled the UV–optical–NIR diffuse Galactic light in the Perseus field using *Euclid* VIS and NISP, CFHT *ugriz*, GALEX FUV/NUV, IRIS 100 μm, and *Planck* dust radiance. Our main results are:

(1) The Perseus DGL/cirrus SED rises from the UV to a broad maximum around the r, i, and I_E bands, and declines toward the NIR. Its overall shape is well reproduced by dust scattered light with a radiative transfer model. **Cirrus is less prominent in the *Euclid* NISP bands owing to two factors: the shallower surface brightness limits and the intrinsically fainter NIR scattered light.** We find no clear evidence for extended red rmission, while the $\sim$20% H_E excess is consistent with an additional contribution from small grains.

(2) The broadband SED constrains the dust albedo ω more strongly than the scattering asymmetry g. The inferred ω is $\simeq 0.42$–0.54 in the UV, $\omega \simeq 0.56$–0.65 across the optical bands, and $\omega \simeq 0.5$–0.6 in the NIR, broadly consistent with previous measurements and Milky Way dust models. The inference depends on the treatment of the FIR normalization: the dynamic I_{100} solution is closer to the D03/WD01 $R_V = 3.1$ model, while the fixed-I_{100} solution is lower, indicating that the dust heating prescription affects the inferred albedo. Measurements across different Galactic sightlines and illumination geometries will be needed to constrain the scattering phase function more directly.

(3) Optical–NIR colors provide useful diagnostics to distinguish Galactic cirrus from diffuse stellar emission. For cirrus, optical colors alone may overlap with intermediate-age or metal-poor stellar populations, while adding the *Euclid* NIR bands produces a clearer separation. In Perseus, $I_E - J_E$ versus $g - r$ gives the clearest distinction between cirrus and the BCG+ICL or evolved stellar population models.

The Perseus field illustrates that with deep imaging, Galactic cirrus is both a foreground that can limit low surface brightness measurements and a resolved tracer of interstellar dust. With *Euclid*, both aspects can be studied using the same data. Extending this analysis across the Euclid Wide Survey will make it possible to study how dust scattering properties and cirrus colors vary across different Galactic environments in a broader picture of the diffuse ISM throughout the Milky Way.

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with the Galaxy Evolution Explorer, obtained from the MAST data archive at the Space Telescope Science Institute, which is operated by the Association of Universities for Research in Astronomy, Inc., under NASA contract NAS5–26555. Based on observations obtained with *Planck* (http://www.esa.int/Planck), an ESA science mission with instruments and contributions directly funded by ESA Member States, NASA, and Canada. This research has made use of the NASA/IPAC Infrared Science Archive, which is funded by the National Aeronautics and Space Administration and operated by the California Institute of Technology.

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Appendix A: Processing of diffuse emission

This appendix details the processing procedures summarized in Sec. 2.5. The overall goal is to extract diffuse Galactic cirrus light while removing compact sources, foreground light, extended stellar halos, ICL, and localized residuals from the *Euclid*, CFHT, and GALEX images.

Appendix A.1: Compact emission separation using `DeNeb`

We extracted the diffuse component of the *Euclid* and CFHT images using `DeNeb`, a software package designed to separate extended ('nebulous') emission from compact sources such as stars and galaxies *skew*(Bertin et al., in prep.). `DeNeb` uses a convolutional neural network trained on images of nebulous emission to separate compact emission from diffuse backgrounds. It has previously been applied to isolate brown dwarfs and planetary-mass objects in LDN 1495 (Bouy et al. 2025), and to study the young star cluster RCW 36 in the Vela molecular cloud (do Brito do Vale et al. 2026). These applications demonstrate its good performance in regions where compact objects are embedded in structured diffuse backgrounds. Detailed photometric performance and completeness tests can be found in do Brito do Vale et al. (2026).

For each image, `DeNeb` produces a compact-emission component and a diffuse emission component. We use the diffuse products from a `DeNeb` v19 run for the VIS and NISP images and for the *ugriz* images. Extended stellar and galaxy halos, faint diffuse galaxies, ICL, and some instrumental artifacts, however, may remain in these products and are treated separately below.

Appendix A.2: Morphological filtering for blob removal

Although `DeNeb` performs well in separating compact emission such as stars and distant galaxies, the extended-emission products still contain localized residuals. These residuals are mainly associated with bright stars, stellar halos of galaxies, **tidal streams around galaxies,** diffuse galaxies, faint stars/galaxies near bright sources, intragroup light, and instrumental artifacts. This is expected because `DeNeb` was not specifically optimized for low surface brightness structures in deep *Euclid* and CFHT imaging. Future dedicated runs of `DeNeb` will improve the separation of such diffuse components.

We therefore applied a morphological filtering method introduced in Liu et al. (2025a). We used the Rolling Hough Transform (RHT; Clark et al. 2014) to identify locally coherent structures characteristic of Galactic cirrus. Compact or blob-like features that remain were detected and added to the mask. We adopted an RHT kernel diameter of 36″ for CFHT and *Euclid*/NISP and 18″ for *Euclid*/VIS, where the finer resolution allows a smaller filtering scale. Features detected above 3σ were masked and subsequently infilled using `maskfill` as described in Appendix B. This step is intended to remove localized residuals while retaining the extended filamentary and patchy structure of the cirrus.

Appendix A.3: Foreground light removal

Zodiacal light is one of the dominant contributors to the optical and NIR sky observed from space (Euclid Collaboration: Borlaff et al. 2022; Windhorst et al. 2022). Its surface brightness depends on wavelength, ecliptic coordinates, the position of the observer, and the time of observation. Across the angular scale of the Perseus field, this foreground is expected to vary smoothly. We modeled this component with `ZodiSURF` (O'Brien et al. 2026), an optical-to-NIR zodiacal light model developed from the SKYSURF measurements (Windhorst et al. 2022). SKYSURF compiles archival HST data from over 200 000 images to measure the sky surface brightness over a wide range of wavelengths and sky positions. `ZodiSURF` extends the Kelsall zodiacal light model (Kelsall et al. 1998) with wavelength-dependent constraints on the scattering properties of interplanetary dust. Details about `ZodiSURF` are referred to O'Brien et al. (2026).

For each *Euclid* and CFHT observation, we generate a 2D zodiacal light model at the corresponding sky position and observing time. We used an Earth-based observer for CFHT and the Earth–Sun L2 geometry for Euclid. The model spectrum is integrated through each filter response to produce a band-averaged zodiacal light map, which is then subtracted from the corresponding science image.

For GALEX, we used the diffuse UV foreground estimates from Murthy (2014). Airglow is removed from both FUV and NUV, while zodiacal light is additionally removed from NUV; its contribution in FUV is negligible. The tabulated foreground estimates are interpolated across the GALEX coverage and weighted by the exposure contributing to each position before subtraction.

Appendix A.4: BCG and ICL subtraction

The dominant contaminant for the cirrus in the Perseus field is the combined light of the BCG NGC 1275 and the ICL. We remove this component from the optical and NIR images before measuring the cirrus. In the followings, "BCG+ICL model" refers to this joint diffuse light template; we do not attempt to decompose the BCG and ICL into separate components.

For the *Euclid* bands, we adopted the BCG+ICL isophote models from Kluge et al. (2025). The VIS model was constructed with elliptical isophotes after accounting for large-scale background structure and Galactic cirrus using the CFHT *i*-band image, a scaled WISE 12 μm dust template, and a first-order 2D gradient. Because cirrus is much fainter in the NISP bands, the NISP models were derived directly from the corresponding *Euclid* images. The construction of these models is described in detail by Kluge et al. (2025).

For the CFHT bands, we used the VIS BCG+ICL model as a template and allowed its amplitude to vary with both wavelength and radius. The ellipse geometry is fixed to the VIS model, while a band-dependent coefficient $s_\lambda(a)$ accounts for radial color variations in the BCG+ICL, where a is semi-major axis distance from NGC 1275. The coefficient is constrained over $20 < a < 100$ kpc, where ICL has high S/N and is less contaminated by cirrus. **We allowed $s_\lambda(a)$ to vary smoothly over this range based on a de Vaucouleurs law ($\propto a^{1/4}$), and remain constant outside this range.**

We fit the CFHT BCG+ICL scaling simultaneously with *Planck* dust radiance and a first-order spatial plane. These additional terms prevent Galactic cirrus and residual large-scale background structure from being absorbed into the BCG+ICL color scaling; they were not subtracted from the CFHT images at this stage. We did not apply an ICL subtraction to the GALEX images, because this component is expected to be negligible for the present analysis. NGC 1275 and its immediate surroundings are instead conservatively masked in both UV bands.

The accuracy of the recovered diffuse emission ultimately depends on the separation between BCG+ICL light and Galactic cirrus (see discussion in Sec. 7.2). Future work will target more accurate separation of extragalactic emission from cirrus.

Appendix A.5: Bright halos, residual sources, and artifacts

After subtracting the BCG+ICL component, we identified extended halos and other source residuals that remained in the images. Candidate features were detected at several spatial scales by running `SExtractor` (Bertin & Arnouts 1996) on images rebinned by factors of 2, 4, 8, and 16. We retained features detected on at least two consecutive scales and used the original images and source catalogs to associate detections with known objects. Most of the resulting features were halos of bright cluster galaxies or optical ghosts and stellar halos in the CFHT images.

Where possible, we modeled the extended light with elliptical isophotes using `photutils` (Bradley et al. 2016). The fits were designed to describe the smooth outer halo rather than the inner source structure, which has already been handled by `DeNeb`. Sources for which a stable halo model could not be obtained were masked instead. Additional mask growth was applied around particularly bright stars, galaxy associations, and visually identified residuals. Strong detector artifacts, most notably in the CFHT z image, were also masked and excluded from the analysis.

Because `DeNeb` was not applied to the GALEX data, sources were detected down to S/N of 2 and masked directly in the FUV and NUV images. These masks were supplemented by visually inspected regions around bright objects. The resulting UV source masks were combined with the NGC 1275 mask and the GALEX coverage masks in the analysis.

Appendix B: Mask-infilling implementation

The combined source and artifact masks left empty regions in the images. Continuity in the diffuse emission is preferred to avoid introducing artifacts during convolution and for visualization. The nominal DGL measurements used observed pixels only; the mask effect on structural analysis requires further validation. We constructed infilled image products using a two-stage procedure: a local infilling using median filtering, and a global infilling based on neural field approach. The mask map constructed for each band was divided into 8-connected components. A component was passed to local filling only if the following conditions were fulfilled: (1) its equivalent radius and its largest bounding-box dimension were no greater than twice the kernel size (2) the maximum distance of its pixel to a valid pixel was no greater than the kernel size, and (3) it was surrounded by at least five valid pixels.

Small mask components with sufficient surrounding information were filled with `maskfill` v1.1.1 (van Dokkum & Pasha 2024), using iterative median reconstruction with an 11×11 kernel. The algorithm was applied only for inpainting, and the unmasked pixels remained unchanged.

Larger or more complex masks were reconstructed using a coordinate-based neural-field model fitted independently in each band. The neural field combines information over several spatial scales and is used where local interpolation alone does not provide adequate support. The predicted intensity is written as

$$I_{\rm pred}(x, y) = I_{\rm backbone}(x, y) + I_{\rm residual,NN}(x, y). \tag{B.1}$$

The backbone is a mask-normalized multiscale Gaussian estimate constructed from original valid pixels. Gaussian scales of 32, 64, and 128 pixels were combined according to their local valid-pixel support using the kernel $G(\sigma)$,

$$B_\sigma(x, y) = \frac{G(\sigma) * [M_{\rm valid}(x, y) \cdot I(x, y)]}{G(\sigma) * M_{\rm valid}(x, y)}, \tag{B.2}$$

where $M_{\rm valid}$ is the mask map that contains only originally valid pixels. Previously infilled values therefore do not contribute to the backbone.

The residual field was represented by a multilayer perceptron operating on Fourier-encoded coordinates. The coordinate basis consisted of x, y, $x + y$, and $x - y$, each expanded with sine and cosine functions at eight logarithmically spaced frequencies extending to 16 cycles per tile following the approach in Tancik et al. (2020). The network contained three hidden layers with 64 units per layer and SiLU activations. Its output layer was initialized to zero so that training started from the smooth backbone.

The residuals were centered and scaled before fitting. For the optimization, we adopted `AdamW` with a learning rate of 0.002 and a batch size of 4096. Original valid pixels were always eligible for residual training. Small-mask pixels filled by local median filtering were included in the training samples to improve continuity, but pixels inside large masks and invalid regions were not used as training data. Each 4800×4800 image was divided into 16 tiles with 384-pixel overlaps. Overlapping predictions were combined with smooth weights. The full-field runs produced finite predictions for all target pixels, while pixels outside the mask remained equal to the input image.

The implementation was tested on cutout regions using artificial circular and elliptical holdouts. On matched holdouts, the neural field performed well from visual recovery and based on root mean square errors. However, its primary purpose in this work is to provide continuous predictions across large masks rather than to accurately recover pixel-level cirrus structures in every band.

The reconstruction could be considered as an interpolation constrained by the multi-scale pixel correlation. The band-independent reconstructed models do not enforce physical color consistency. Caution must be taken that small-scale structures could be suppressed or spuriously introduced inside large mask.

Appendix C: Analytic details of the radiative transfer model

In Sec. 4, we introduced the radiative transfer framework used to interpret the measured DGL SED. Here we describe the calculation in more detail. Our aim is to predict the scattered intensity along the Perseus line of sight for a specified dust model, while treating the illumination and attenuation by the Galactic stellar and dust distributions in a self-consistent way. We follow the plane-parallel single-scattering framework developed by Brandt & Draine (2012).

Appendix C.1: Vertical distributions and incident radiation

We approximate the Milky Way as an infinite plane-parallel disk in which both the stellar emissivity and dust density vary only with height z above the Galactic midplane. This idealized geometry captures the first-order dependence of the scattered light intensity on Galactic latitude and on the vertical distributions of stars and dust (Jura 1979).

We describe the stellar luminosity density as $q_\lambda(z) = L_\lambda \varphi_s(z)$ where L_λ is the intrinsic luminosity per unit area of the stellar disk and $\varphi_s(z)$ describes its normalized vertical distribution. We adopt two exponential stellar components,

$$\varphi_s(z) = f_1 \frac{\exp(-|z|/h_{s,1})}{2h_{s,1}} + (1 - f_1) \frac{\exp(-|z|/h_{s,2})}{2h_{s,2}}, \tag{C.1}$$

with $h_{s,1} = 300\,\mathrm{pc}$, $h_{s,2} = 1350\,\mathrm{pc}$, $f_1 = 0.9$. The two scale heights approximately represent the thin- and thick-disk stellar

populations. This parameterization was based on observational constraints on the vertical stellar distribution of the Milky Way (e.g., Gilmore & Reid 1983; Binney & Merrifield 1998). The profile is normalized such that $\int_{-\infty}^{+\infty} \varphi_s(z)\, dz = 1$.

For the dust distribution, we adopt a Gaussian vertical profile for the number density,

$$n_{\rm H}(z) = n_{\rm H,0} \exp\left(-\frac{z^2}{2h_d^2}\right), \tag{C.2}$$

with $h_d = 250$ pc. A Gaussian profile is motivated by observations of the vertical distribution of Galactic interstellar material (Malhotra 1995; Nakanishi & Sofue 2003).

Instead of introducing the midplane normalization $n_{\rm H,0}$ as an additional free parameter, we set it through the adopted vertical optical depth. The optical depth at V band is given by

$$\tau_{V,0} = C_{{\rm ext},V} \int_0^{+\infty} n_{\rm H}(z)\, dz. \tag{C.3}$$

We set $\tau_{V,0} = 0.15$ following the fiducial normalization used by Brandt & Draine (2012), which is the optical depth through one Galactic half-plane from the midplane to $z = +\infty$. In the ideal plane-parallel limit, the corresponding **line-of-sight** optical depth at Galactic latitude b scales as

$$\tau_V(b) = \tau_{V,0} \csc|b|. \tag{C.4}$$

At wavelength λ, we define the cumulative vertical optical depth above height z as

$$\tau_\lambda(z) = \int_z^{+\infty} C_{{\rm ext},\lambda} n_{\rm H}(z')\, dz', \tag{C.5}$$

where $C_{{\rm ext},\lambda}$ is the wavelength-dependent extinction cross-section per H nucleon provided by the dust model.

We consider a stellar sheet at height z_s and a dust grain at height z. For a source element separated from the grain by projected distance R in the Galactic plane, the physical source-to-grain distance is $r = \sqrt{R^2 + (z - z_s)^2}$. The optical depth encountered by a photon traveling from the source to the grain is

$$A_\lambda(z, z_s, R) = |\tau_\lambda(z) - \tau_\lambda(z_s)| \frac{r}{|z - z_s|}. \tag{C.6}$$

After integrating over the projected source radius, the contribution from a stellar sheet can be written as

$$K_\lambda(z) = \frac{1}{2} \int_{-\infty}^{+\infty} \varphi_s(z_s) E_1\left(|\tau_\lambda(z) - \tau_\lambda(z_s)|\right)\, dz_s, \tag{C.7}$$

where $E_1(x)$ is the exponential integral: $E_1(x) = \int_x^\infty e^{-t}/t\, dt$.

The attenuated mean stellar intensity at height z is then:

$$J_\lambda(z) = \frac{L_\lambda}{4\pi} K_\lambda(z). \tag{C.8}$$

For the spectral shape of the illuminating field, we adopt the local interstellar radiation field of Mathis et al. (1983). MMP83 describes the radiation field measured locally near the Solar neighborhood, rather than the intrinsic luminosity of the Galactic disk before attenuation. We determine L_λ by requiring the attenuated model to reproduce the MMP83 intensity at the midplane:

$$J_\lambda(0) = J_{\lambda,{\rm MMP83}}, \tag{C.9}$$

which gives

$$L_\lambda = \frac{4\pi J_{\lambda,{\rm MMP83}}}{K_\lambda(0)}. \tag{C.10}$$

As a result, the model reconstructs the intrinsic stellar emissivity required to reproduce the adopted local ISRF after propagation through dust extinction.

Appendix C.2: Scattering geometry and integrated DGL

Next we calculate the amount of light scattered into the line of sight, where anisotropy of the scattering becomes important. Previous measurements have shown that the ratio of NIR DGL to 100 μm emission varies systematically with Galactic latitude, providing direct evidence that anisotropic scattering and Galactic geometry affect the observed DGL (Sano et al. 2016a). Sano & Matsuura (2017) showed explicitly that a plane-parallel Milky Way model can reproduce much of this latitude dependence.

For a grain at z and a stellar sheet at z_s, we define

$$u = \frac{R}{|z - z_s|}. \tag{C.11}$$

The cosine of the physical scattering angle ξ, defined between the photon incident direction and the outgoing direction toward an observer at Galactic latitude b, is given by:

$$\cos\xi = \frac{u\cos|b|\cos\theta - {\rm sgn}(z - z_s)\sin|b|}{\sqrt{1 + u^2}}. \tag{C.12}$$

The ${\rm sgn}(z - z_s)$ occurs because stars above and below the grain illuminate it from different directions.

For the nominal model, we use the Henyey–Greenstein (HG) phase function (Henyey & Greenstein 1941):

$$\phi_\lambda(\cos\xi) = \frac{1}{4\pi} \frac{1 - g_\lambda^2}{\left(1 + g_\lambda^2 - 2g_\lambda\cos\xi\right)^{3/2}}, \tag{C.13}$$

with $\int_{4\pi} \phi_\lambda(\cos\xi)\, d\Omega = 1$. The scattering asymmetry parameter $g_\lambda = \langle\cos\xi\rangle$ describes the mean scattering anisotropy: $g = 0$ corresponds to isotropic scattering and $g = 1$ corresponds to complete forward scattering. The HG form has been widely adopted in DGL studies because it provides a simple one-parameter description of anisotropic dust scattering. However, it has been shown that HG phase function poorly describes the scattering anisotropy in the UV and NIR (Draine 2003b; Baes et al. 2022).

For a vertical optical-depth separation $\Delta\tau_\lambda = |\tau_\lambda(z) - \tau_\lambda(z_s)|$, we combine attenuation of the incident radiation and angular redistribution into the scattering kernel

$$\mathcal{K}_{\lambda,{\rm sca}}(\Delta\tau_\lambda, u, b) = \int_0^\infty \frac{u\, du}{1 + u^2} \exp\left[-\Delta\tau_\lambda\sqrt{1 + u^2}\right] \times \int_0^{2\pi} \phi_\lambda(\cos\xi)\, d\theta. \tag{C.14}$$

The single-scattering emissivity per steradian toward the observer is then given by:

$$j_{\lambda,{\rm sca}}(z, b) = n_{\rm H}(z) C_{{\rm ext},\lambda} \omega_\lambda \frac{L_\lambda}{4\pi} \int_{-\infty}^{+\infty} \varphi_s(z_s) \mathcal{K}_{\lambda,{\rm sca}}(\Delta\tau_\lambda, u, b)\, dz_s\,. \tag{C.15}$$

In this expression, $C_{{\rm ext},\lambda}$ sets how frequently the incident photons interact with the dust, ω_λ sets what fraction of those interactions result in scattering, and g_λ controls how strongly the scattered photons are redirected toward or away from the observer.

The scattered photon is additionally attenuated between the grain and the observer. For an observer in the Galactic midplane,

$$\tau_{\lambda,{\rm obs}}(z, b) = |\tau_\lambda(z) - \tau_\lambda(0)| \csc|b|. \tag{C.16}$$

The total scattered light intensity is therefore

$$I_{\lambda,{\rm sca}}(b) = \csc|b| \int_0^{+\infty} j_{\lambda,{\rm sca}}(z, b) \exp[-\tau_{\lambda,{\rm obs}}(z, b)]\, dz. \tag{C.17}$$

The b-dependence of $I_{\lambda,\mathrm{sca}}$ in Eq. (C.17) is important considering the relative illumination from the stellar disk, the scattering angles, and the attenuation through line of sight. This latitude dependence has been observed in the optical (Zhao et al. 2024) and NIR (Sano et al. 2016a) DGL and discussed in Sano & Matsuura (2017).

Appendix C.3: Absorbed energy and FIR normalization

To compare the radiative transfer prediction with our measured $\nu I_\nu / I_{100}$ SED, we also calculate the total energy absorbed by the same dust distribution. In radiative equilibrium, this absorbed energy provides the power budget reradiated in the infrared. The absorbed intensity along the sightline is

$$I_{\mathrm{TIR}}(b) = \frac{\csc|b|}{4\pi} \int_0^{+\infty} dz \int_0^{\infty} d\lambda\, n_{\mathrm{H}}(z) C_{\mathrm{ext},\lambda}(1-\omega_\lambda) 4\pi J_\lambda(z). \tag{C.18}$$

Following Brandt & Draine (2012), we use the DL07 dust model to convert I_{TIR} to 100 μm intensity for the normalization (Eq. 5): $\langle \nu I_\nu \rangle_{100\mu\mathrm{m}} = 0.52 I_{\mathrm{TIR}}$. Alternatively, we consider a fixed normalization using the observed 100 μm intensity (see Sec. 5.2).

Appendix C.4: Bandpass integration and numerical implementation

The model is evaluated monochromatically, whereas the DGL measurements are broadband quantities. We therefore integrate each prediction through the corresponding filter response curve. For a band X with throughput $T_X(\lambda)$,

$$\langle I_\nu \rangle_X = \frac{\int I_\nu(\lambda) T_X(\lambda) d\lambda/\lambda}{\int T_X(\lambda) d\lambda/\lambda}. \tag{C.19}$$

The quantity compared with the observed SED is $\langle \nu I_\nu \rangle_X / I_{100}$, determined from the filter throughput curve.

For the calculation, we integrate the vertical distributions from −3 to +3 kpc. The vertical grid sampling is verified to converge within 1%. The model scattering spectrum spans from 0.10 to 10 μm on logarithmic wavelength grid.

The scattering integration uses a logarithmically spaced u grid between 10^{-4} and 10^4. The u grid and azimuthal grid sampling are verified to converge within 1%, where the difference is mainly from UV bands and is negligible in NIR. We tabulate the scattering kernel on a grid of $\Delta\tau_\lambda$ for interpolation during the radiative transfer calculation.

Our model remains intentionally idealized. It assumes plane-parallel stellar and dust distributions and only treats single scattering. It does not include the radial structure of the Galactic disk, spiral arms, localized illuminating sources (e.g., OB stars), spatial variations in grain properties, or a complete treatment of non-equilibrium dust emission. The latter may become relevant at the longest wavelengths: both observational and modeling studies have suggested that NIR DGL can contain contributions in addition to scattered starlight (Sano et al. 2015; Sano & Matsuura 2017). The purpose of the present model is therefore not to reconstruct the full 3D radiation field toward Perseus, but to provide a controlled physical framework in which the dependence of the measured DGL SED on ω_λ, g_λ, extinction, and geometry can be quantified. This framework also forms the basis of the dust parameter inference described in Sec. 5.

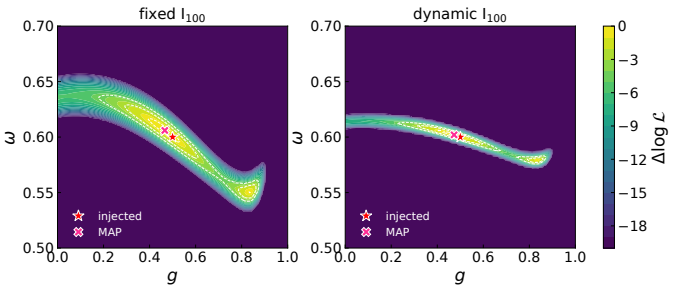


Fig. D.1: Likelihood surface for recovery of the constant-input model with fixed and dynamic I_{100}, **showing the degeneracy between ω and g**. The color scale shows the likelihood relative to its maximum. The star marks the input values, $(\omega, g) = (0.6, 0.5)$, and the cross marks the maximum found on the numerical grid.

Appendix D: Validation of the dust parameter inference

Before applying the inference to the Perseus measurements, we tested whether it can recover known dust scattering properties from synthetic DGL broadband photometry. These experiments have two purposes: first, they provide a check that the radiative transfer calculation, bandpass integration, likelihood, and posterior sampler are mutually consistent; second, they show which combinations of dust parameters can be distinguished by broadband photometry with wavelength coverage similar to our data. We consider two input recovery experiments. In each experiment, we generated a synthetic SED from a specified dust model and performed inference with the framework used for the observations. **To separate degeneracies in the model itself from the effect of measurement noise**, we reduced mock uncertainties to 25% of those of the measured Perseus SED. These tests therefore represent a deliberately higher S/N case than the real data. Because the mock data contain only the modeled scattered light component, H_{E} is treated as a detection rather than as an upper limit.

Appendix D.1: Recovery of constant scattering parameters

We first consider a simple model, with wavelength-independent $\omega = 0.6$ and $g = 0.5$ from the UV through the NIR. This test removes the additional freedom associated with wavelength-dependent dust properties and allows us to examine directly the degeneracy between the parameters. We started with evaluating the likelihood directly on the 2D (ω, g)-plane. A coarse grid over the full physical range was first used to identify all relevant modes, followed by a finer grid calculation around the likelihood maxima. Further grid refinement produces negligible changes in the recovered likelihood. Figure D.1 shows the likelihood relative to its maximum for the fixed and dynamic normalizations. **The maximum a posteriori (MAP) estimates are close to the injected values.**

We then sample the same likelihood with MCMC, using 24 walkers evolved for 1500 steps and initialized around the grid maximum. The recovered parameters are $(\omega, g) = (0.60, 0.48)$ for fixed I_{100} and $(\omega, g) = (0.60, 0.47)$ for dynamic I_{100}, close to the input values in both cases. The albedo is constrained relatively well. The asymmetry parameter is less well determined, and the fixed normalization likelihood also contains a secondary region near $(\omega, g) = (0.56, 0.83)$. This secondary mode indicates that even with reduced uncertainties a more forward scattering

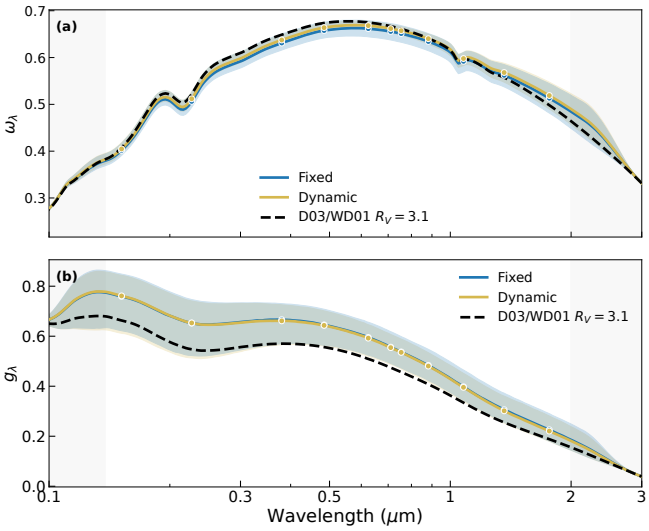


Fig. D.2: Recovery of the fiducial D03 albedo and asymmetry curves from a synthetic broadband DGL SED. The input curves are compared with the posterior envelopes for the fixed- and dynamic-I_{100} treatments. The return to D03 outside the fitted wavelength range is imposed by the tapering and does not represent information supplied by the synthetic data.

phase function can be partially compensated by a lower albedo and produce a similar broadband SED.

Appendix D.2: Recovery of the fiducial D03 model

For the second experiment, we generate the synthetic SED from the wavelength-dependent D03 $R_V = 3.1$ albedo and asymmetry curves and recover it using the deformation model of Sec. 5.1. The input parameter vector is therefore simply $(a_\omega, t_\omega, c_\omega, a_g, t_g) = (0, 0, 0, 0, 0)$. For this validation only, we use broad uniform priors on all deformation coefficients to replace the model-envelope prior on g introduced in Appendix D.3. This is because the purpose of this test is to measure what information is supplied by the synthetic DGL likelihood itself. Centering a g prior on D03 would otherwise favor the known input by construction. We sample each normalization branch with 40 walkers for 2600 steps, discard the first 800 steps, and thin according to the measured autocorrelation. The resulting recovery is shown in Figs. D.2.

For both normalization treatments, the input albedo and asymmetry curves lie within the recovered 68% posterior envelopes over the wavelength range constrained by the synthetic photometry. However, there is an offset in g_λ. The corresponding shift in a_g and t_g is accompanied by a smaller offset in the albedo parameters, consistent with the ω–g degeneracy already visible in the first experiment.

We therefore regard this test as evidence that the implementation is moderately capable of recovering the input model at the precision allowed by the broadband SED. However, there is no clear evidence that all the deformation coefficients are unbiased or well determined. Combinations of ω_λ and g_λ could produce very similar broadband photometry, which requires additional information to break the degeneracy.

In conclusion, the two recovery experiments establish the main limitation that guides our real data inference. The broadband DGL photometry contains appreciable information on the albedo, but considerably less on the scattering asymmetry. Allowing a flexible, weakly constrained g_λ therefore could introduce bias and correlation that are not clearly supported by the data. This motivates the low dimension model-envelope prior described in Appendix D.3.

Appendix D.3: Construction of the model-envelope prior on g

The recovery tests in Appendices D.1 and D.2 show that broadband photometry along a single line of sight does not provide strong constraints on the scattering asymmetry. With broad priors, a wide range of $\boldsymbol{\theta}_g$ can reproduce very similar SEDs, where changes in g can be compensated by changes in ω. Since the real data have larger uncertainties than the recovery tests, we regularized the deformation in g with a weak prior in the inference of the real data.

The prior was built from the spread among a set of Milky Way dust models: the D03/WD01 $R_V = 3.1$ model and the $R_V = 4.0$ and $R_V = 5.5$ models (Draine 2003b), the ZDA04 BARE–GR–S model (Zubko et al. 2004), the DustEM model (Compiègne et al. 2011), and the Astrodust model (Hensley & Draine 2023). Each model was projected in the same latent space used in the inference following the inverse of Eq. 9. The prior is centered on the fiducial D03/WD01 $R_V = 3.1$ model, corresponding to $\boldsymbol{\theta}_g = (0, 0)$.

We described the resulting envelope with a 2D Student-t distribution with four degrees of freedom. Its scale parameters were chosen so that all models lie within the joint 68% region. This corresponds to 1σ half-widths of approximately 0.4 in a_g and 0.2 in t_g. We adopted a diagonal scale matrix, so the prior does not impose a correlation between the deformation coefficients. The Student-t distribution provides broader tails than a Gaussian prior, and therefore allows the posterior to explore values beyond those spanned by the reference models.

This model-envelope prior is intended only to regularize a parameter combination that the broadband SED constrains weakly. If the posterior remains close to the prior, it does not imply that the data independently favor the D03 model. Instead, it indicates that broadband photometry alone provides limited information on g. This is different from the behavior of the albedo, for which the recovery tests show that ω_λ is strongly constrained by the broadband SED itself.